LUMINESENSE

A Capstone Project Proposal

Presented to the Faculty of the

College of Information Technology

University of Negros Occidental – Recoletos

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

by

Alexandra Shierlmole S. Ballesteros

John Gemar E. Entapa

Athea Grace O. Talanquines

2025

LIST OF FIGURES

1. LumineSense General Features. 15

2. LumineSense Logical Prototype. 26

3. LumineSense Physical Prototype. 27

4. Default Classroom Design of the LumineSense System. 30

5. LumineSense Decomposition Chart. 44

SINCE 2010
Bachelor of Science in
INFORMATION
TECHNOLOGY

LIST OF TABLES

1. Comparative Matrice of Related Applications. 6

2. Hardware Components Requirements. 18

3. Hardware Development Requirements. 19

4. Minimum Device Specifications. 20

5. Software Specifications. 22

6. Firmware Libraries Specifications. 24

7. Standards for Classroom Dimensions. 29

8. Types of Sensors and Dependencies. 31

9. Defined Set of Hand Gestures. 33

10. Connectivity Requirements. 34

11. Power Sources of Main Components. 35

12. Power Requirements of Main Components. 36

13. Occupancy Detection Activity Diagram. 48

14. Gesture Recognition Activity Diagram. 51

15. Timetable-Based Scheduling Activity Diagram. 55

16. Web-based Monitoring Activity Diagram. 59

17. Video Feed Display Activity Diagram. 62

18. Automation and Regulation Activity Diagram 65

19. Classroom Variability Activity Diagram. 68

20. User Access Control Activity Diagram. 71

TABLE OF CONTENTS

LIST OF FIGURES i

LIST OF TABLES ii

I. INTRODUCTION 1

1.1 Purpose 3
1.2 Technical Review of Related Systems 5
1.2.1. Sensor-based Lighting Detection 8
1.2.2. Centralized Management Dashboard 9
1.2.3. Schedule Management System 9
1.2.4. Gesture Control for Lighting 10
1.3 Project Scope 10
1.3.1. Motion Detection and Control System 11
1.3.2. Prevention of Energy Waste through System-based Automation 11
1.3.3. Schedule-based Lighting Management 12
1.3.4. Secure and Authorized Use 12
1.3.5. Scalability of Project Prototype 13

II. SYSTEM SPECIFICATIONS 14

2.1. System Perspective and General Perspective 14
2.1.1. Lighting Automation and Control 15
2.1.2. Schedule-Based Usage Monitoring 16
2.1.3. Gesture-Controlled Lighting Commands 16
2.1.4. Energy Consumption Analytics 16
2.2. Operating Environment 17
2.2.1. Hardware Requirements 17
2.2.2. Software Tools 20
2.2.3. Connectivity Requirements 24
2.3. Design and Implementation Constraints 25
2.3.1. Logical and Physical Prototype Design 25
2.3.2. Ideal Classroom Setting and Lumens Variability 28
2.3.3. Ideal Physical Design 29
2.3.4. Sensor Dependencies and Mitigation 30
2.3.5. Gesture Controls 31
2.3.6. Connectivity Dependence 33
2.3.7. Power Sources and Consumption of Components 34
2.4. Assumptions and Dependencies 36
2.4.1. Assumptions 36
2.4.2.1. Constant Network Connectivity in Educational Settings 37

2.4.2.2. Available and Uninterrupted Power 37
2.4.2.3. Classroom and Environmental Suitability 37
2.4.2.4. Capabilities of Arduino Components 38
2.4.2.5. Shelf Life or Longevity of Components 38
2.4.2. Dependencies 39
2.4.2.1. Functioning Components 39
2.4.2.2. Active and Secure Server 40
12.4.2.3. Account Validation and Authentication 41

III. SYSTEM FEATURES 41

3.1. System Decomposition 41
3.2. System Functionalities 43
3.2.1. Occupancy Detection 45
3.2.2. Gesture Recognition 48
3.2.3. Timetable-Based Scheduling 51
3.2.4. Web-based Monitoring 55
3.2.5. Video Feed Display 59
3.2.6. Automation and Regulation 62
3.2.7. Classroom Variability Options 65
3.2.8. User Access Control 68

VI. PROJECT MANAGEMENT 72

6.1. User Classes and Characteristics 72
6.1.1. Facility Managers and Staff 72
6.1.2. Students 73
6.1.3. Faculty Members 73
6.2. Product Feasibility and Assessment 74
6.2.1. Technical Feasibility 74
6.2.1.1. Core Functionalities 74
6.2.1.2. Data Availability and Integration 75
6.2.2. Market Feasibility 76
6.2.3. Operational Feasibility 77
6.3. Time Management 78
6.4. Communication, Coordination, and Team Composition 79
6.4.1. Communication 79
6.4.2. Coordination 79
6.4.3. Team Composition 80
6.4.3.1. Writer 80
6.4.3.2. Developer 81
6.4.3.3. Designer 81

GLOSSARY 83

APPENDICES 87

A. Capstone Project Proposal 87
B. PERT Table. 89
C. Gantt Chart. 90
D. Minutes of the Capstone Project Proposal Defense. 91
E. System Logo 94
F. Team Logo 95

REFERENCES 96

CURRICULUM VITAE 100

1.1 Purpose 3
1.2 Technical Review of Related Systems 5
1.2.1. Sensor-based Lighting Detection 8
1.2.2. Centralized Management Dashboard 9
1.2.3. Schedule Management System 9
1.2.4. Gesture Control for Lighting 10
1.3 Project Scope 10
1.3.1. Motion Detection and Control System 11
1.3.2. Prevention of Energy Waste through System-based Automation 11
1.3.3. Schedule-based Lighting Management 12
1.3.4. Secure and Authorized Use 12
1.3.5. Scalability of Project Prototype 13

II. SYSTEM SPECIFICATIONS 14

2.1. System Perspective and General Perspective 14
2.1.1. Lighting Automation and Control 15
2.1.2. Schedule-Based Usage Monitoring 16
2.1.3. Gesture-Controlled Lighting Commands 16
2.1.4. Energy Consumption Analytics 16
2.2. Operating Environment 17
2.2.1. Hardware Requirements 17
2.2.2. Software Tools 20
2.2.3. Connectivity Requirements 24
2.3. Design and Implementation Constraints 25
2.3.1. Logical and Physical Prototype Design 25
2.3.2. Ideal Classroom Setting and Lumens Variability 28
2.3.3. Ideal Physical Design 29

2.3.4. Sensor Dependencies and Mitigation 30
2.3.5. Gesture Controls 31
2.3.6. Connectivity Dependence 33
2.3.7. Power Sources and Consumption of Components 34
2.4. Assumptions and Dependencies 36
2.4.1. Assumptions 36
2.4.2.1. Constant Network Connectivity in Educational Settings 37
2.4.2.2. Available and Uninterrupted Power 37
2.4.2.3. Classroom and Environmental Suitability 37
2.4.2.4. Capabilities of Arduino Components 38
2.4.2.5. Shelf Life or Longevity of Components 38
2.4.2. Dependencies 39
2.4.2.1. Functioning Components 39
2.4.2.2. Active and Secure Server 40
12.4.2.3. Account Validation and Authentication 41

III. SYSTEM FEATURES 41

3.1. System Decomposition 41
3.2. System Functionalities 43
3.2.1. Occupancy Detection 45
3.2.2. Gesture Recognition 48
3.2.3. Timetable-Based Scheduling 51
3.2.4. Web-based Monitoring 55
3.2.5. Video Feed Display 59
3.2.6. Automation and Regulation 62
3.2.7. Classroom Variability Options 65
3.2.8. User Access Control 68

VI. PROJECT MANAGEMENT 72

6.1. User Classes and Characteristics 72
6.1.1. Facility Managers and Staff 72
6.1.2. Students 73
6.1.3. Faculty Members 73
6.2. Product Feasibility and Assessment 74
6.2.1. Technical Feasibility 74
6.2.1.1. Core Functionalities 74
6.2.1.2. Data Availability and Integration 75
6.2.2. Market Feasibility 76
6.2.3. Operational Feasibility 77
6.3. Time Management 78
6.4. Communication, Coordination, and Team Composition 79
6.4.1. Communication 79
6.4.2. Coordination 79

6.4.3. Team Composition 80
6.4.3.1. Writer 80
6.4.3.2. Developer 81
6.4.3.3. Designer 81

GLOSSARY 83

APPENDICES 87

A. Capstone Project Proposal 87
B. PERT Table. 89
C. Gantt Chart. 90
D. Minutes of the Capstone Project Proposal Defense. 91
E. System Logo 94
F. Team Logo 95

REFERENCES 96

CURRICULUM VITAE 100

CHAPTER I

INTRODUCTION

Energy conservation is a concept that appeals to many different types of stakeholders. Its huge effect on addressing climate change and global warming challenges and offering potential to reduce energy prices has led to its widespread practice, development, and discussion in the political, industrial, and governmental sectors. Few studies have addressed this subject in the context of academic institutions, which have one of the biggest effects on energy consumption, primarily facilitated by man's capacity for planned behavior [EE2018]. Due to this, energy becomes an asset that, when mismanaged, costly consequences potentially arise. This concern has continually been remedied, especially through the field of Internet of Things (IoT) technology, where real-time data and predictive analysis are used to optimize energy loads, significantly balancing energy consumption [KK2023].

The United Nations Sustainable Development Goals (SDGs), specifically SDG 7: Affordable and Clean Energy, cover exactly this area of concern, where energy-related concerns are vital. Under this goal, ensuring readily accessible, clean, and affordable energy includes the means of investing in long-term growth, whether it be investing in renewable energy sources or expanding energy efficiency [UN2024]. Nonetheless, energy consumption is a dominant contributor to climate change, especially through the greenhouse gases (GHG) it emits [UN2024]. The conservation of energy could not be stressed enough now that it covers a wide range of industries, including healthcare, manufacturing, as well as the institutional environment.

Because of population growth, the educational sector is expanding rapidly. These schools need to create and use enormous amounts of energy units. The environment will suffer as a result of rising carbon emissions and the depletion of non-renewable energy sources. Furthermore, severe weather conditions significantly raise energy demand. To increase the energy performance of school facilities worldwide, several energy efficiency programs have been devised, particularly for the optimization of energy consumption [AJ2016]. By means of applying IoT, it allows us to manage educational infrastructure and facilities, which in turn has the potential to increase real-time facility maintenance, monitoring, and efficiency. A smarter and more responsive management system is essential in this digital age to provide an efficient and long-lasting teaching and learning process [HI2024]. LumineSense integrates with this platform to provide a seamless approach towards remote, security, and control management of various features, including light controls and consumption maintenance.

LumineSense is an IoT-based smart lighting management system that integrates motion detection, automated controls, and centralized monitoring to optimize energy consumption, especially in classrooms. The system uses Passive Infrared (PIR) sensors and web cameras to detect activity, triggering microcontroller units to activate or deactivate lights via relay switches based on real-time occupancy data. A centralized web-based dashboard provides authorized personnel with monitoring capabilities for energy consumption, lighting schedules, and security alerts during non-operational hours. By automating light control and eliminating human error, LumineSense addresses energy waste in unoccupied rooms while enhancing campus security.

1.1 Purpose

LumineSense aims to be a smart, user-friendly lighting management system tailored for educational institutions. Its goal is to optimize energy consumption through automated, adaptive lighting controls while enhancing the learning environment. LumineSense ensures operational simplicity for facility managers with robust security and detailed analytics, all without requiring deep technical knowledge. By achieving these integrated goals, LumineSense positions itself as a complete solution that tackles the sustainability and operational issues that educational institutions encounter when overseeing classroom lighting systems. The system eventually advances the more general objectives of institutional cost reduction and responsibility for the environment.

1.1.1. Power Consumption Reduction

Through implementation, unwanted power consumption is reduced by using sensors, light bulbs, and a sense of automation. LumineSense addresses energy reduction in educational institutions, where lights are left on in unoccupied spaces, resulting in excessive electricity consumption and increased operational costs. By implementing smart automation, this will significantly reduce the amount of energy that is normally wasted when it is not in use. In the long run, this leads to significant cost savings and more efficient electricity use in educational facilities, especially in classroom settings.

1.1.2. Lighting Management

LumineSense aims to provide centralized, sustainable, and smart control of classroom lights. This is done through multiple control mechanisms which are occupancy detection, gesture controls, and schedule-based automation aligned with class timetables. All lighting control features are combined into a single web-based dashboard, and this

allows administrators to modify lighting schedules across multiple classrooms. Due to this, less manual switching and maintenance are required.

1.1.3. Report Mechanism

LumineSense offers reporting features that record lighting activities and system operations across classes under observation. The system creates reports, occupancy detections, schedule adherence, and system alerts. Administrators may watch when and where lights were turned on, which operate as official records for institutional responsibility. This reporting feature promotes institutional adherence to sustainability policies and operational procedures while guaranteeing openness in energy management methods.

1.1.4. Data Analytics

LumineSense provides simple and accessible data analytics that help administrators understand energy usage patterns immediately. Easy-to-read graphs that display daily and weekly lighting usage across monitored classrooms are used by the system to deliver energy consumption statistics. These graphs are supported by summaries that emphasize total operating hours and estimated energy consumption. Administrators may understand the facility's energy efficiency and identify areas for improvement. This feature encourages an approach of constant enhancement in energy management methodologies while enabling administrators to show observable results by converting energy data into useful information.

1.2 Technical Review of Related Systems

Rainbow8™ Smart Light Emitting Diode (LED) [NA2016], Wireless Sensor Network (WSN)-based Energy Conservation Prototype [MK2016], IoT Smart Lighting over Light Switches [ME2020], IoT Contactless Switch Using Gesture Control [VP2025], and IoT Photovoltaic (PV)-based Smart Lighting System for Classrooms [MS2023] were the existing systems and applications being reviewed had immensely contributed through the conception of LumineSense. Essentially, LumineSense encapsulates remote, security, and control management specifications through IoT technology. In the market, there are many IoT applications for smart lighting industries, especially in light bulbs that offer wireless control, energy efficiency, and a sense of automation, but through these supported systems,

LumineSense is closely aligned with automated light management and control applicable towards particular classroom and educational settings, sustaining the use of energy while contributing towards issues of energy conservation. Each reviewed system provided unique insights into different aspects of the implementation, from gesture-based controls to renewable energy integration, allowing the team to identify both strategies and limitations in existing studies. The analysis revealed that while commercial smart lighting products prioritize consumer experience, they often lack the specialized features required for institutional environments such as schedule-based automation and comprehensive energy reporting.

By synthesizing the strengths of these systems while also addressing their gaps, LumineSense offers a tailored approach that balances automation, user control, and sustainability for educational facilities. In itself, the development process later is administered in alignment with the reviewed systems and answers how it contrasts and compares with the project's scope.

System Criteria	Rainbow8™ Smart LED	WSN-based Energy Conservation Prototype	IoT Smart Lighting over Light Switches for Classrooms	IoT PV-based Smart Lighting System for Classrooms	IoT Contactless Switch Using Gesture Control	LumineSense
Sensor-based Lighting Activation	Wireless Fidelity (Wi-Fi) and Bluetooth (mobile)	Motion (PIR) Temperature (LM35) Light (LDR) ZigBee Transreceiver (Wi-Fi signaling)	Motion (PIR) Temperature (PIR) Relay switches (signaling)	Motion (PIR) Light; detecting illumination and intensity (LDR)	Relay module (signaling)	Motion and temperature (PIR) Relay switches (signaling)
Manage-ment Dashboard	Only location (mobile)	Lighting and appliance control	Reports, lighting, and location management (web and mobile)	Real-time light and energy status (web)	Only output	System status, energy use, trends, short- and long-term summaries
Schedule Manage-ment System	Time and location (mobile)	Pre-defined	Time and location (web and mobile)	Automated tied to the classroom timetable (web)	None	Controllable and automated through timetables
Gesture Control	None	None	None	None	Camera-based	Web camera (webcam)

Table 1. Comparative Matrice of Related Applications.

One device in relation to the management and control present in the study was iLuv's Rainbow8™ Smart LED. With the help of Wi-Fi, this multicolored smart LED light bulb may be controlled from any location where illumination is linked. Compared to incandescent bulbs, which use 1200 kWh of electricity over the same time measurement, this plug-and-play IoT system integrated with the Apple IoT HomeKit platform allows users to save energy by 85%, using only 180 kWh over a period of 20k hours [NA2016]. If further developed, energy-saving smart products will prove to be a solution particularly as a contributing factor to LumineSense.

Like LumineSense, a prototype is essential as a testing ground for feasibility. Another study was conducted regarding its systematized smart home energy conservation using a WSN [MA2016]. The authors formulated the following main components: light, temperature or heat, and motion. In terms of motion, the use of PIR sensors in the study allowed automatic lighting control when motion was detected in a room with low illumination and then turning it off after a set delay when no motion was sensed. In terms of temperature, the study utilized it as a way to monitor environmental conditions, tackling issues in energy savings, and comfort. A Zigbee transceiver was also used to communicate sensor data and control signals between different parts of the sensor system. With this review, LumineSense can capitalize on the use of sensors well.

Moreover, another local study about enclosed devices around light switches is also centered around an Arduino-based smart lighting system [ME2020]. The study integrated the said IoT platform around typical light switches commonly found in classrooms that when a certain schedule has reached its duration, a relay switch component will automatically turn off the light. Additionally, on top of the capability to set schedules, the project also has a report management system wherein information about actions done and electricity consumed per hour is detailed. All of the system's features are also controlled via a dedicated controller application, where users can manipulate lighting controls and view other features, something LumineSense could expound on.

In addition, the IoT-based Contactless Switch using gestures [VP2025] details a touch-free controllability over appliances and lighting. The project was designed to reduce physical contact between shared switches. It uses a camera-based hand tracking system with models from Open Computer Vision Library (OpenCV) and MediaPipe to interpret real-time hand gestures. This

concept is a point of reference for LumineSense by allowing users to override lighting through hand gestures detectable by web cameras.

Furthermore, a more recent study developed a solar-powered classroom lighting system that combines IoT technology with renewable energy sources [MS2023]. Data is transmitted through Wi-Fi to a web-based interface, where users can observe real-time illumination and system performance. It also includes a scheduling feature aligned with classroom timetables to ensure optimal energy use. While the system's strength lies in its integration of solar energy and IoT, LumineSense further extends these features with data-driven reporting and security enhancements beyond lighting automation.

By combining several sensor inputs, such as the PIR, LumineSense builds on existing products like Rainbow8™, which provides energy-efficient smart LED lighting controlled via Apple HomeKit. With the help of this combination, LumineSense may adjust brightness in response to occupancy and natural sunshine, providing a highly flexible, energy-efficient, and intuitive lighting experience in classrooms. Additionally, scheduling, remote control, thorough consumption reporting, and security features are all included in the extensive IoT platform, which offers a single solution tailored to the operational and sustainability requirements of educational institutions. LumineSense offers a new type of smart lighting system that easily finds a balance between comfort, energy efficiency, and campus safety by building on prior prototypes that mostly relied on simple PIR sensor control and scheduling.

1.2.1. Sensor-based Lighting Detection

LumineSense includes a relay for fine-grained lighting control, light intensity sensors for daylight capture, and PIR sensors for real-time occupancy detection. In reaction

to ambient light, its auto-control procedures modify the lighting intensity as well as the on/off states, ensuring that only the appropriate lighting is provided and reducing waste. The system logs abnormalities for diagnostic purposes and has a 10-minute pause to prevent unintentional shutdown during idleness. In comparison with the studies, the common sensor-based detection technologies were primarily PIRs with relay switches and Wi-Fi modules to interconnect such devices.

1.2.2. Centralized Management Dashboard

Occupancy of all rooms, real-time energy consumption, light state, and daylight conditions are centrally monitored through the web-based dashboard. With summaries, energy consumption and savings metrics are displayed and made available for download, giving facility managers useful information. Compared to other systems the developers reviewed, each lacked analytics, some offering basic scheduling, and others not saving any data. LumineSense addresses all these limitations by tracking occupancy, light levels, energy use, and schedules, and saving all data in a structured query language. This allows administrators to view real-time room status, generate reports, and review energy savings with reliable data. With regards to historical data, information can be summarized by month, weeks, or any timeframe to give more scrutiny over data-driven decisions. In essence, the thorough analytics and reporting provided by LumineSense are especially tailored to the objectives and operational schedules of educational institutions.

1.2.3. Schedule Management System

LumineSense enables administrators to create custom lighting schedules based on original class timetables, operational hours, and even weather or daylight conditions.

Sensor-based control can be supported or overridden by schedules as needed, allowing the best possible lighting for learning. In contrast to simple time or pre-defined triggers present in the studies, intelligent technology in LumineSense is prioritized to accommodate the complex patterns of occupancies and actual room usage, allowing for automated, energy-efficient campus operations.

1.2.4. Gesture Control for Lighting

To provide a more interactive lighting experience, LumineSense incorporates the existence of a web camera capable of detecting user gestures for basic lighting control. Instead of relying solely on physical switches or mobile applications, the system interprets movements to trigger lighting adjustments automatically. This feature promotes ease of use and a touch-free environment, which is especially useful in shared classroom settings. Unlike other systems that limit user input to manual or app-based control, LumineSense introduces camera-assisted functionality that enhances both convenience and system responsiveness.

SINCE 2010
Bachelor of Science in
INFORMATION
TECHNOLOGY

1.3 Project Scope

With automated monitoring and management, LumineSense enables educational institutions to automate lighting control and improve energy efficiency. The application delivers automatic motion-based lighting and security monitoring features, giving facility administrators operational control. Without constant manual intervention or extensive technical knowledge of IoT systems, the main goal is to help administrators and facility managers make informed decisions and optimize energy usage while ensuring efficient operation, whether classrooms are occupied or

empty. By considering this general system functionality, further understanding of certain project boundaries and scope details respective to the main project objectives is important.

1.3.1. Motion Detection and Control System

The motion detection allows the system to instantly modify lighting output based on occupancy and daylight levels. By recognizing infrared radiation from heat and in motion, the PIR sensors identify occupancy. When motion is detected, the lighting is automatically activated. To ensure that lighting adjustments are carried out precisely based on combined sensor inputs, relay modules enable either independent or collective control of bulb zones. This enables flexibility amongst authorized personnel, granting convenience and efficiency in comparison to manual lighting procedures.

1.3.2. Prevention of Energy Waste through System-based Automation

The automation operates without physical switches or manual intervention, relying entirely on sensor-driven logic and programmed schedules where the PIR sensor automatically deactivates lights after detecting no movement for a configurable duration (default 10 minutes), eliminating the common scenario where lights remain on in empty classrooms through a logical process that operates continuously during defined operational hours with all activation and deactivation events logged to the local database for analysis and reporting. The system tracks energy consumption patterns by recording the duration lights remain active and correlating this data with occupancy detection events and scheduled class times, enabling facility managers to access real-time energy usage statistics, daily and weekly consumption summaries, and comparative analytics through the web dashboard that highlight potential areas for further optimization. By providing

quantifiable metrics on energy savings achieved through automation, LumineSense assists classroom maintenance teams in monitoring productivity and justifying continued investment in smart building technologies.

1.3.3. Schedule-based Lighting Management

Classes are planned at specified times throughout the day, and educational facilities follow set schedules. Administrators typically need a mechanism to pre-configure lighting behavior to fit the known occupancy patterns of most schools. By giving authorized staff, the ability to create unique lighting plans for various rooms based on class schedules and business hours, LumineSense offers schedule management. The scheduling method provides an alternative to developing time-based lighting rules that support or replace motion-based management to further improve this feature.

1.3.4. Secure and Authorized Use

LumineSense is designed to support educational institutions by automating and monitoring classroom lighting. The system restricts access through role-based permissions, with administrators granted full control over scheduling, energy analytics, and system configuration through the web dashboard. Faculty members benefit from the automated environment without requiring direct system access, though these users can utilize gesture controls for manual overrides during class sessions. With respect to the Data Privacy Act of 2012, the system does not record any unnecessary or personal information. Only occupancy-related data required for lighting automation are collected, and these data are used solely to trigger and manage the automated functions performed by the system, in compliance with data privacy and protection standards [NP2012]. This authorization

structure ensures operational security while maintaining ease of use, with all user actions logged for accountability and system transparency.

1.3.5. Scalability of Project Prototype

Time-bound limitations determine the overarching progress of the project. Arduino components are required to be outsourced while integrating software tools, both new-found and existing, requires understanding of key semantics. By accounting the IoT and software-based frameworks to be implemented, the LumineSense system is expected to handle accurate software-side data and display appropriate system-wide information with the physical prototype acting as a bridge between the two. The concept of prototyping the IoT system so that it aligns well with the information to be presented is important, which has main reliance over scalability. For that reason, the prototype used in the system is intended to be scalable for the aformentioned factors all the while aligning with project functionalities and objectives.

SINCE 2010
Bachelor of Science in
INFORMATION
TECHNOLOGY

The chapter emphasizes that academic institutions and significant electricity consumers are frequently disregarded despite their high energy consumption, even though energy-saving initiatives are prominent in the governmental and industrial sectors. To address this gap, LumineSense is introduced as an IoT-based smart lighting management system that leverages motion detection, automated scheduling, and a centralized dashboard to reduce energy waste and improve efficiency. Related technologies and systems establish the foundation for LumineSense, demonstrating how it integrates and enhances features all the while accounting for the project scope, assumptions, and dependencies laid out in the system.

CHAPTER II

SYSTEM SPECIFICATIONS

This chapter outlines the general features of the project, covering foundational aspects of the study and further examining concepts introduced in the previous chapter. This IoT project enables energy conservation through automation and control monitoring, allowing for the control of energy overconsumption via the application's restrictions and other features. Furthermore, the project team aims to delineate the manual lighting it takes on usual classroom settings, purposefully eliminating the need for physical management, as well as addressing sustainability in light energy conservation.

2.1. System Perspective and General Perspective

The LumineSense system is designed to automate and optimize classroom lighting operations through an integrated IoT-based platform. This functions as a way to remotely utilize and interact with various features available throughout the system. Additionally, by using sensors to detect motion and gestures systematically control and automate lights when needed, the system will be able to monitor energy consumption in the long run. In general, LumineSense can benefit in the factors of saving electricity, lowering cost, and making lighting control effective and efficient, normalizing the sustainable use of energy of light in classroom settings. As a holistic view, LumineSense paves the way for IoT to become a modular management application that integrates hardware and software frameworks for novel solutions of recurring lighting issues.

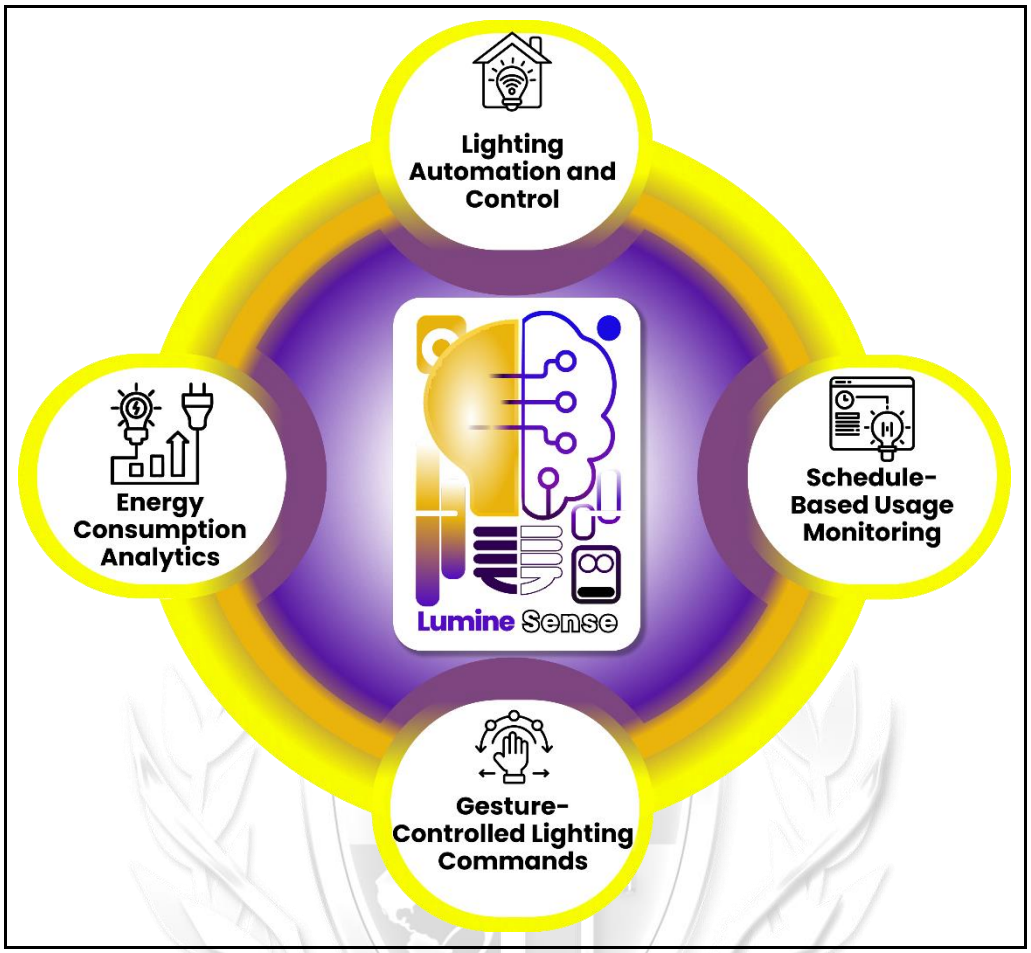


Figure 1. LumineSense General Features.

Bachelor of Science in INFORMATION TECHNOLOGY

2.1.1. Lighting Automation and Control

The IoT system automates classroom lighting through motion and light sensors. When movement is detected, primarily centered around the PIR sensor, the light turns on, and when the room is empty, it will automatically turn off after a short delay set within the program. By using the Arduino controller and wireless modules, the LumineSense system allows remote control of light through a dashboard. Usage restriction is also present in the system, specifically for hours outside of class hours, wherein users will be able to set

specific timetables or schedules, which the auto-detection should work, signifying that entry is prohibited for energy conservation.

2.1.2. Schedule-Based Usage Monitoring

The LumineSense system includes a scheduling feature that aligns lighting operations with a class timetable, available through an integrated web-based dashboard. The system will automatically follow the set schedule by activating and deactivating lights according to daily class and maintenance routines, especially during in-class hours and special classroom occupancies. With regards to room occupancy requests, the users will be able to set a schedule available through the website, and then the system will follow suit.

2.1.3. Gesture-Controlled Lighting Commands

The gesture control feature allows users to easily regulate classroom lighting by using pre-programmed hand gestures that are picked up by the built-in web camera. These motions improve convenience in classrooms by enabling touchless commands like turning lights on or off and changing overall classroom brightness by specifying which row of lights to be turned on. In addition to its automatic lighting features, the project's gesture recognition technology offers an interactive interface that gives consumers direct control as needed.

2.1.4. Energy Consumption Analytics

As stated previously, the system automates classroom lighting through motion and light sensors. The system keeps track of energy usage in a classroom where the system

resides, showing a report of real-time data presentable through the website dashboard. Through this feature, administrators can make data-driven decisions to improve energy efficiency, justify maintenance activities, and evaluate the long-term impact of smart lighting in a classroom setting.

2.2. Operating Environment

The operating environment of LumineSense refers to the hardware and software components required for its optimal performance. It includes sensors, lighting control modules, cameras, and a central server or computer that manages data processing and user interaction. The software environment comprises the control dashboard, database system, and communication protocols that connect devices and transmit real-time data. Additionally, LumineSense may rely on third-party libraries or APIs for data analytics and alert notifications. Overall, the system's operating environment is an integrated setup of hardware and software designed to ensure stable operation, accurate monitoring, and efficient energy management.

2.2.1. Hardware Requirements

. The LumineSense embedded system will be supported by various sensors, as the developers have stipulated. First and foremost, the PIR sensor, specifically the HC-SR501 PIR motion sensors detect occupancy by sensing motion in the room, providing a baseline for turning the lights on or off based on presence [LM2018a]. To control the switching of connected lighting circuits, the team will use a four-channel 5V relay module [CO2021], allowing the Arduino to safely turn different light zones on or off. A 5V 5A Adapter Power

Supply will provide the stable power necessary for the components. An ESP32-DevKitC V4 Wi-Fi Module will enable wireless connectivity for network communication between the microcontroller and the dashboard or other control interfaces [MR2021]. To ensure voltage stability by providing a constant 3.3V supply required by components like the ESP32, an AMS1117-3.3V and ACS712 Current Sensors are needed [AI2023]. To measure voltage and energy flowing in and out of the bulbs for lighting automation and measurement, the AC 22KW Digital Voltage Meter and the PZCT-02 Current Transformer (CT) Sensors are required as performed by studies attempting smart light automation and measurement. A 4-Channel Bi-directional Logic Level Shifter safely converts logic voltage levels between 5V and 3.3V systems, critical for interfacing components that operate at different voltages [CE2022]. DS3231 Real-Time Clock (RTC) Module keeps accurate time for scheduling lighting events and logging activities [LM2018b].

Hardware Component	Specification
Motion Sensors	HC-SR501 PIR
Relay Module	4-Channel 5V
Power Supply	5V 5A Adapter
Wi-Fi Module	ESP32-DevKitC V4
Voltage Regulators	AMS1117-3.3V and ACS712 Current Sensor
Digital Voltage Meter	AC 22KW Digital Voltage Meter
Current Transformer (CT) Sensor	PZCT-02 CT Sensor
Logic Level Shifter	4-Channel Bi-directional
RTC Module	DS3231 Real-Time Clock

Table 2. Hardware Components Requirements.

To ensure efficient development and prototyping accuracies, reliable market-ready hardware components were selected for both performance and efficiency. The Arduino

Mega 2560 serves as the central controller, coordinating system logic and sensor integration [TE2018]. An 830-point breadboard with jumper wires enables flexible, solderless prototyping connections between components. Resistors and capacitors provide signal conditioning, pull-up or pull-down configurations, filtering, and power smoothing throughout the circuit. In the prototyping phase, the bulbs are an important factor to consider, as the system will use nine pieces of LED 3W Mini Bulbs for prototyping. A Micro SD Card 8GB Class 10 MicroSD Card offers local data storage for logs, configuration files, and energy consumption reports, which is essentially data storage as a whole.

Hardware Component	Specification
Main Controller	Arduino Mega 2560
Breadboard	830 points
Jumper Wires	Male-Male, Female-Male
Resistors	10kΩ, 220Ω, 470Ω (various values)
Capacitors	100µF, 10µF, 1000µF (various values)
Prototype Bulbs (9 pcs)	LED 3W Mini Bulbs
MicroSD Card	8GB Class 10

Table 3. Hardware Development Requirements.

In terms of the web dashboard, it requires minimal performance outputs for proper information and data to be disseminated and transformed across the hardware to software framework the system has. Therefore, the developers also determined the minimum requirements for a laptop personal computer (PC) to present the web dashboard the LumineSense system has to ensure proper display of system operations. A standard

Windows 10 or higher suffices, as gesture processing occurs on edge hardware rather than the host PC, ensuring smooth dashboard rendering even during peak occupancy logging. Integrated graphics handle video streams of at least 2.5 Gigahertz (GHz) or higher are adequate for classroom monitoring, aligning with prototype constraints while enabling scalability for institutional deployment. Additionally, when locally initializing LumineSense, 2 to 3 or more Gigabytes (GB) of storage are necessary for baseline data displays and smooth management in the side of the system's dashboard. A built-in webcam is also important to enable contactless lighting controls via user's response and confirmation.

Hardware (PC) Component	Specification
Operating System	Windows 10 or higher
Processor Speed	At least 2.5 GHz or higher
Random Access Memory (RAM)	8 GB or higher
Secondary Storage	2-3 GB of Free Storage
External Components	Webcam

Table 4. Minimum Device Specifications.

2.2.2. Software Tools

The team employed various development tools for the project, ranging from collaboration tools to languages essential for IoT and other programming frameworks. The primary Integration Development Environments (IDEs) are Arduino IDE and Visual Studio Code. Arduino IDE is a comprehensive, lightweight editor for building applications across numerous embedded systems [AR2025]. Meanwhile, Visual Studio Code is a

powerful, all-rounder source code editor for various languages, with in-app downloadable extensions for expanding development, including embedded systems [VS2025].

The LumineSense dashboard is developed and integrated on a comprehensive local server that combines cross-platform Apache, MySQL, Perl, and PHP (XAMPP) [XA2025]. Through this, the team can establish a local database architecture for storing lighting schedules, logs, and energy consumption reports, streamlining the development process by enabling simultaneous front-end interfacing and back-end operations.

In terms of the embedded system being used, C++ is utilized for programming the Arduino MCU because of its low-level hardware control capabilities and effective memory management, which are crucial for real-time sensor processing. On the other hand, languages like the Hypertext Markup Language (HTML) with Cascading Style Sheet (CSS), Bootstrap, and Asynchronous JavaScript (AJAX) are collectively used for the web-based dashboard, creating visually cohesive and responsive interfaces that adapt across devices, and enabling dynamic real-time updates.

SINCE 2010
INFORMATION
TECHNOLOGY

For searching for credible sources and related systems, Google and ProQuest search engines were used, providing the team with a wide range of technical references and the most recent IoT implementations and innovations. Collaborative documentation and version tracking is done through Microsoft Word and GitHub, where the team can work on the same document while tracking changes, revisions, and versions efficiently. For communications, the most available and easy-to-use platforms, such as Microsoft Teams and Discord, were used, enabling quick submission and transmission of messages, files, and on-time project updates. This makes real-time collaboration and updates possible.

Category	Specification	
IDEs	Arduino IDE, Visual Studio Code	
Server Framework	XAMPP	
Data Sending	MySQL	
Programming Languages	Back-end	Front-end
	C++	HTML, CSS, Bootstrap, JavaScript/AJAX
Design Tools	Figma	
Operating System	Windows 10+	
Search Engines	Google, ProQuest	
Communication Platforms	Microsoft Teams, Discord	
Collaboration Platforms	Microsoft Word (OneDrive)	
Version Control Platform	GitHub	

Table 5. Software Specifications.

The firmware environment present in the LumineSense system relies on essential frameworks and libraries that enable seamless communication between hardware components and software interfaces. The following libraries provide the system necessary protocols and functions for network communication, sensor data processing, and computer vision features.

Network connectivity and data transmission are made possible through the numerous built-in and third-party libraries the ESP32 Wi-Fi Module has. This Arduino component has a built-in WiFi.h library allows it to connect to wireless networks and handle its credentials [AR2025]. By sending HTTP queries to the web server, the HTTPClient.h library enables the MCU to transmit commands and sensor data from the dashboard. While ArduinoJson facilitates effective parsing and creation of JavaScript Object Notation (JSON) data structures and API connection, WebServer.h allows the Wi-

Fi module to operate as a local web server for direct configuration and monitoring [AR2025].

For the DS3231 RTC Module, it primarily utilizes the RTClib library. This framework provides comprehensive functions for managing time-based features such as getting, setting time, and managing alarms. This library is crucial for the schedule management system, enabling accurate time-based lighting control in accordance with classroom timetables.

For data management and storage, the local server will be centered around the SD Card Module which employs two built-in Arduino libraries: SD.h for file operations including reading, writing, and managing files on the microSD card, and the SPI.h for Serial Peripheral Interface (SPI) interaction between the SD card reader and the Arduino [AR2025].

The gesture recognition feature leverages two prominent computer vision frameworks: Open Source Computer Vision (OpenCV) Library and MediaPipe. The OpenCV framework provides comprehensive image processing capabilities, including frame capture from the IP camera with visual feedback rendering [VS2023]. MediaPipe, developed by Google, offers pre-trained machine learning models specifically designed for real-time hand tracking and gesture recognition [VP2025][JX2025]. As demonstrated in recent studies, the integration of these two frameworks enables accurate detection of hand landmarks and gesture classification, making touchless lighting control both practical and reliable [VP2025]. Together, these libraries form the software backbone that translates

physical hand movements into actionable lighting commands within the LumineSense system.

Component	Firmware Specifications
ESP32 Wi-Fi Module	Libraries: WiFi.h (built-in); HTTPClient.h; WebServer.h; ArduinoJson.h
DS3231 RTC Module	Library: RTClib
SD Card Module	Libraries: SD.h (built-in); SPI.h (built-in)
Gesture Recognition (from IP Camera)	OpenCV, MediaPipe

Table 6. Firmware Libraries Specifications.

2.2.3. Connectivity Requirements

The application LumineSense uses requires a stable internet connection from development to users, especially in educational institutions, where connecting IoT devices to a centralized server is most important. A minimum of 5-10 megabits per second (Mbps) is enough to support real-time transmission of multiple feeds, from sensors to cameras being used, especially Wi-Fi connectivity between the Arduino MCU and the local server running XAMPP. Data transmitted will be stored in the SD Card Module, where the local server will access recorded data or reports. This makes network access important for authorized personnel to closely monitor the web-based dashboard across various devices. The real-time functionality of LumineSense, which includes automated sensor responses, live dashboard updates, and instant alert notifications, would be severely hampered without sufficient bandwidth and reliable network infrastructure, undermining the system's primary value proposition of responsive, intelligent lighting management.

2.3. Design and Implementation Constraints

As detailed in the previous sections, this project implements various features, typically from control and management of lighting systems, that distinguish it from other IoT lighting-based applications. However, integrating these features in the long run comes with constraints that influence how the software and hardware integrations are either limited or detailed, depending on the scope the project tackles. With consideration of these key metrics, the team is limited to only the following criteria, especially in the nature of being an embedded system and the components, it requires to functionally implement LumineSense.

2.3.1. Logical and Physical Prototype Design

The prototype design of LumineSense is engineered as a scaled-down demonstration system that validates the technical and operational feasibility of the IoT system without requiring full classroom deployment. Rather than implementing the system in actual classroom setting, the prototype serves as a proof-of-concept that validates the feasibility of the project. This is to ensure that this controlled environmental approach allows the system to practically test and refine each integration of the embedded components while proving the system's core capabilities as a manageable, transportable format. In future development, the prototype the project offers can be made as a scalable module which leaves an open opportunity for testing and identifying the capabilities and limitations of the prescribed project prototype in line with its main objectives.

The physical prototype consists of two separate display boards: one that houses the Arduino system and another that features nine LED bulbs arranged in three rows of three

bulbs each. The purpose of this arrangement is to mirror common ceiling lighting configuration found in most classrooms. Moreover, this enables demonstration of the zone-based lighting control which is performed and executed by sensor-based responses and gesture commands. The bulbs are then connected to relay modules controlled by the Arduino Mega 2560, with PIR sensors and an IP camera strategically positioned both to detect motion and temperature respectively.

Essential hardware components detailed in Section 2.2.1 are integrated into the prototype including the DS3231 RTC module to aid for time-based scheduling, the ESP32 Wi-Fi module for network connectivity, and the SD Card Module for local data logging.

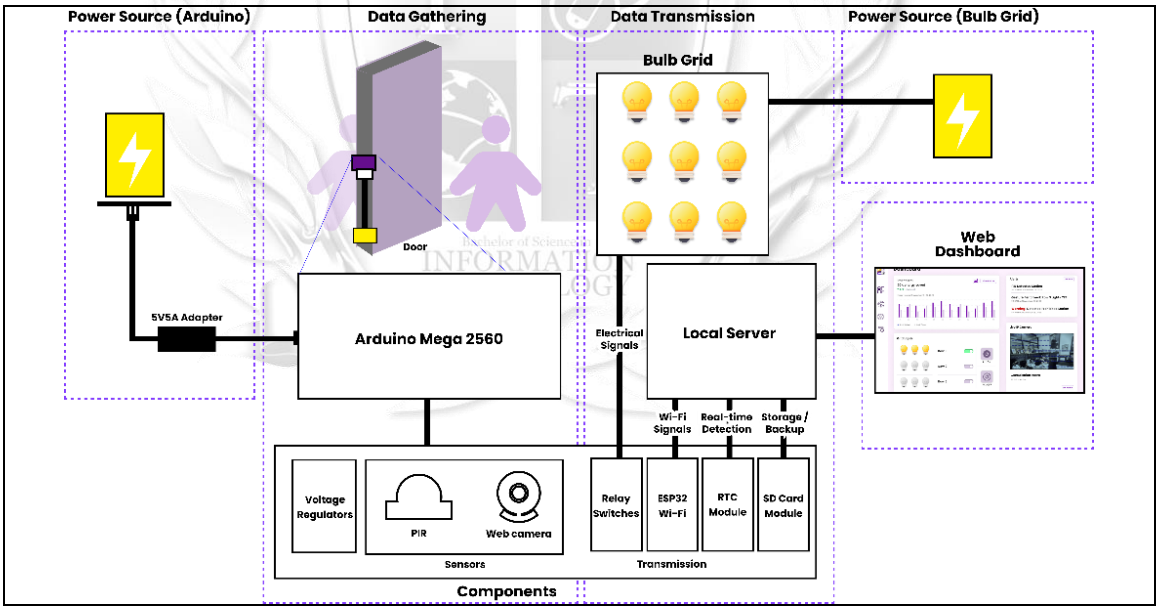


Figure 2. LumineSense Logical Prototype.

Figure 2 presents the Logical Prototype of LumineSense's architecture and data flow of the LumineSense smart lighting system. The Arduino Mega 2560, which serves as

the main controller that receives data from sensors and manages the overall lighting operation. The system uses motion based on occupancy and gesture-related sensors for web cameras as inputs, while relay switches control the lighting mechanisms of the bulb grid. Other supporting modules include the ESP32 Wi-Fi module, RTC module, and SD card module which handle connectivity, time-based scheduling, and data logging. Voltage meters and current sensors are also included to monitor power usage and power inflow and outflow. Overall, the figure illustrates how LumineSense is logically connected via sensors modules and other Arduino components towards lighting outputs which enable automated, scheduled, and energy-efficient lighting.

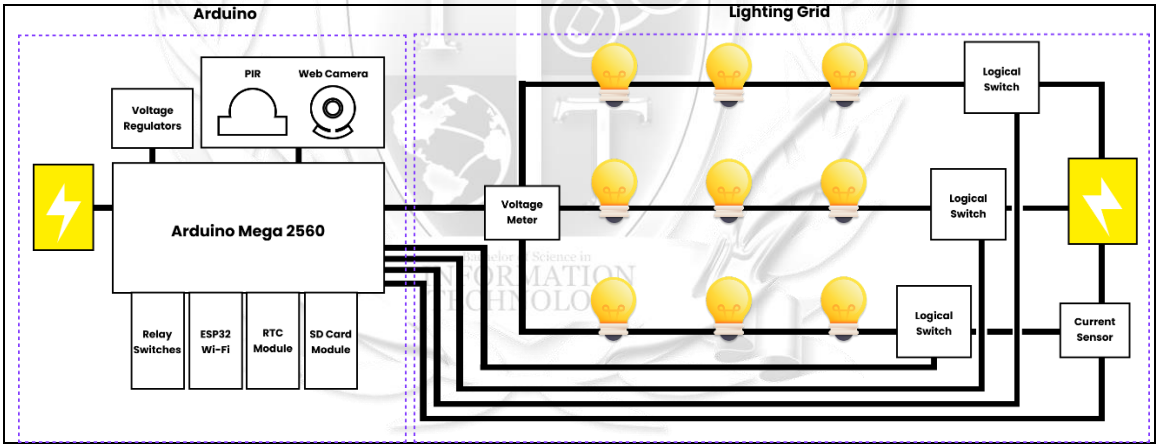


Figure 3. LumineSense Physical Prototype.

The physical design prioritizes portability and self-containment, enabling the prototype to be transported for project defense and demonstrations while operating independently through a dedicated power supply. In addition, two power sources are inevitably present in the prototype: one for the Arduino components and the other for the

9 prototype bulbs. This modular construction allows component testing and troubleshooting, with boards providing visual access to internal wiring, sensor placement, and relay connections.

This controlled demonstration environment proves the system's capability to detect occupancy, respond to ambient light conditions, recognize predefined gestures, follow schedule-based automation, and log energy consumption data which all is attained without requiring installation in an active classroom. By validating sensor accuracy, testing response times, and refining gesture recognition algorithms in this setting, the team collects meaningful performance data that supports the system's viability for eventual real-world implementation in educational institutions.

2.3.2. Ideal Classroom Setting and Lumens Variability

The specifications of which the system is installed are defined in this section, which is important in determining the recommended layout in account for various components contradicting with the physical dimensions of the classroom. The developers have defined an ideal classroom setting in reference to the Department of Education (DepEd) Order No. 64 established on 2017 with respect to the IoT components being used in the project. In DepEd school buildings, classrooms are specified to have a standard size of 7 meters (m) in width and 9 m in length, or 63 square meters (sq. m) in totality [DE2017]. In terms of the variability of dimensions, in essence, the sizes of the classroom, the developers defined these metrics as shown in Table 7 to be part of the options for the users may be adhered to when using the classroom. As a standardized basis, there should be at least 1-2 sq. m per

student depending on the common capacity of the room [VD1985]. The prescribed lighting level for all types of general classrooms is a range between 300 and 500 lumens per sq. m. (lux) [EU2021] with a minimum of 250 lux to achieve modern standards for lumens per watt in an ideal classroom [DE2017].

Classroom Type	Typical Dimensions	Total Area (sq. m.)	Common Capacity	Required lux (lumen/sq. m.)
Small	7m x 7m	49 sq. m.	30-35 students	300 lux (or 3000 lumens)
Medium (Standard)	7m x 9m	63 sq. m.	45-46 students	300-500 lux (3000-5000 lumens)
Large	9m x 10m above	90 sq. m.	30+ (with equipment)	500+ lux (5000+ lumens)

Table 7. Standards for Classroom Dimensions.

2.3.3. Ideal Physical Design

With respect to the ideal classroom dimensions and prescribed lux standardized previously, the physical design of LumineSense adheres to scalable configurations that accommodate different classroom sizes while maintaining optimal illumination standards. For medium-sized (the standard) classrooms measuring 7 m x 9 m (63 sq. m.), the system employs a 3x3 grid arrangement of nine LED bulbs as demonstrated in the prototype, providing coverage for 45-46 students with illumination ranging from 3000-5000 lumens. This configuration ensures uniform light distribution across the space while allowing zone-based control through relay modules, enabling selective activation of bulb rows based on occupancy patterns or user preferences.

For small classrooms (7 m x 7 m, 49 sq. m.), a 2x2 grid of four LED bulbs suffices to meet the minimum 300 lux requirement, delivering approximately 3000 lumens for 30-35 students. Large classrooms exceeding 90 sq. m. require 4x4 or larger grid arrangements to maintain the 500 and above lux standard necessary for spaces accommodating specialized equipment or higher occupancy. The modular design allows institutions to adapt the system's physical layout to existing infrastructure, with PIR sensors positioned centrally for maximum coverage and IP cameras mounted at optimal vantage points to capture gesture inputs without obstruction.

Figure 4. Default Classroom Design of the LumineSense System.

2.3.4. Sensor Dependencies and Mitigation

The sensors implemented are inherent to intrinsic detection and accuracy limits, specifically the PIR sensor and the built-in web camera. Therefore, standardizing accuracies and dependencies in this project requires consistent metrics, calibrations, and interdependencies present in the components being used. The HC-SR501 PIR sensors are sensitive to environmental deficiencies, including temperature, field of view, direct sunshine, and other impediments [AK2020]. Additionally, these sensors could only detect within a certain radius, about 30 feet [VE2025]. Cameras are also susceptible to accuracy considerations when using frameworks like MediaPipe [AH2022][PT2025]. Thus, to reduce errors and maximize efficiency, strategic sensor placement, calibration, and environmental pre-testing are crucial.

Sensor Type	Range	Description	Dependencies	Mitigation	Threshold
PIR	30 feet	Measures IR radiation or heat emitted by warm objects	Clear line-of-sight, temperature stability, RTC for scheduling	Environmental pre-testing, calibration	80% and higher
Image sensing (from web camera)	Effective up to 7 meters	Vision-based gesture recognition using camera	Network bandwidth, lighting conditions, processing speed	Dashboard control fallbacks	70% and higher

Table 8. Types of Sensors and Dependencies.

2.3.5. Gesture Controls

There are several studies the developers have found that support the concept wherein gestures allow control mechanisms over lighting, specifically over LED bulbs. With the interconnection of OpenCV and MediaPipe, relay switches have made this possible with the incorporation of an Arduino MCU over LEDs [VP2025]. In the same recognition software, this time with the incorporation of an ESP32 Wi-Fi module, wireless gesture control and management over LEDs is possible [JX2025].

Forementioned IoT studies prove that camera-based gesture controls are possible over lighting mechanisms, given that proper technologies bridge the two together. By supporting these claims, the developers have defined a set of gestures as shown in Table 3 based on MediaPipe's "canned gestures" that are predefined and pretrained in its default gesture recognizer machine learning model [GA2025].

Table 9 enumerates the following gestures to be implemented in the system, each having its own response suitable for lighting automation controls. Six gestures (Open Palm, Closed Fist, Pointing Up, Victory/Peace, "I Love You", and Thumbs Up) are present in MediaPipe system by default which allows the programmers to allocate respective responses for the IoT framework. Users will also need to confirm the gesture by performing Thumbs Up or by selecting a dashboard-based fallback in order for inaccuracies and accidental activations be delimited. Nonetheless, the remaining gestures allow the user to automate the LumineSense lighting system: performing the Open Palm turns all the lights on and Closed Fist will turn all lights off, Pointing Up, Victory/Peace, and "I Love You" turns the first/second/third row of lights respectively, either on/off depending on the current

state of the bulb. Finally, the Thumbs Up gesture mentioned previously for the sake of confirmation activation.

Hand Gesture	MediaPipe Canned Gestures	Response
	Open Palm	Turn all lights on
	Closed Fist	Turn all lights off
	Pointing Up	1st row of lights turned on/off
	Victory/Peace	2nd row of lights turned on/off
	“I Love You”	3rd row of lights turned on/off
	Thumbs Up	Gesture confirmation

Table 9. Defined Set of Hand Gestures.

2.3.6. Connectivity Dependence

LumineSense relies on Wi-Fi connectivity facilitated via the ESP32 module, which is necessary for data transfer in real-time between sensors, the MCU, and the dashboard housed on a server. Due to this system's full functionality, ranging from motion detection to live energy monitoring, proper management is contingent on a stable and continuous network connection as shown in the constraints detailed in Table 10. Each component in the LumineSense system is either part of the embedded IoT or as one of software functionalities, with connections relying towards general-purpose I/O (GPIO) and even in local data transfers. Thus, data transmission between entities in the IoT components may be impacted by traffic or blocked ports, primarily because most academic buildings share the same network infrastructure. Therefore, the strength and stability of the network are essential for guaranteeing the operational functionality of LumineSense.

Component	Connectivity Type	Bandwidth Requirement	Dependency
ESP32 Wi-Fi	Wi-Fi	5-10 mbps minimum	Critical
Arduino Mega 2560	Serial Communication	N/A	Critical
Web Camera	Wi-Fi/Ethernet	2-5 mbps	High
Web Dashboard	Internet/Local Network	1-4 mbps	High
XAMPP	Local Network	Stable connection	Critical
PIR Sensors to MCU	GPIO	N/A	Critical
Relay Modules	GPIO	N/A	Critical

Table 10. Connectivity Requirements.

2.3.7. Power Sources and Consumption of Components

LumineSense has two major power sources to be determined in the development phase, one for the Arduino IoT sensors and connections and the other for lighting grid. It is important to understand the difference between the two power sources as the consumption between the structures is varying and collision of electrical differences affect inflow and outflow of energy to be recorded accurately. The presence of multiple sensors and modules also leaves the IoT system vulnerable to power imbalances which could potentially cause data inconsistencies. Thus, it is significant that the division between the Arduino and the bulbs are highlighted and identified. Furthermore, by clarifying the contrasting sources between the main components, it is imperative that the power emitted by the sources is enough to support the sub-components it is connected with.

Main Component	Related Components	Power Sources
Arduino MCU	o PIR Sensor o RTC Module o SD Card Module o ESP32 Wi-Fi o Host device's webcam (gesture recognition)	5V 5A Adaptor
Lighting Grid	o 9 Bulbs o Logical Level Shifters o Current Sensor (data transmission) o Voltage Meter (data transmission)	220 V AC

Table 11. Power Sources of Main Components.

When considering the development of the prototype, levels of power flowing from component to component varies and it is important to ensure that all elements have

sufficient inflow of electricity in order to function properly. Based on manufacturer specifications [TE2018][LM2018a][CE2022], the following components must function well which in turn require sufficient power supplied to them because the main LumineSense functionalities rely on its constant presence and delivery. With an improper or imbalanced supply of energy, data logging and displays are prone to inaccurate readings, especially during the transmission of data as it goes from the hardware to the software side.

Component	Minimum Voltage Requirements (V)	Power Consumption (W)	Primary Feature(s)
Arduino Mega 2560	5 or higher	1.10	Processing
PIR Sensor	3.3 to 5	0.00	Occupancy Detection
Logical Level Shifters	3.3 to 5	1.50	Processing and Logical Controls
RTC Module	3.3 to 5	0.00	Scheduling and Real-time Control
SD Card Module	5	0.5	Data Logging

Table 12. Power Requirements of Main Components.

Bachelor of Science in INFORMATION TECHNOLOGY

2.4. Assumptions and Dependencies

LumineSense integrates several features for intelligent lighting control and management, and the following purposes align well with the project's goals. Its performance depends on specific technical and environmental factors that ensure proper operation. These include the reliability of network connectivity, sensor accuracy, hardware functionality, and server availability. The following assumptions and dependencies define the required conditions and limitations under which LumineSense can operate effectively as an IoT-based embedded system.

2.4.1. Assumptions

While it is important to ensure that functionalities are fully conceptualized in the long run, various constraints may be encountered that potentially deter the scope and intention of the project. Assumptions are also documented to become an important asset in the development process wherein boundaries are acknowledged in which parts the project should be limited and abiding towards such assumptions. The team has identified the following scenarios in which the LumineSense system follows through.

2.4.2.1.Constant Network Connectivity in Educational Settings

The presumed academic classroom where the LumineSense is supposed to be installed is assumed to have a reliable and secure local network that can accommodate real-time sensor detection and data transmission in IoT application, in which the web dashboard should display proper and accurate information. For this reason, the system requires constant access to Wi-Fi or Internet connectivity. Additionally, the classroom's IT infrastructure is assumed to have minimal latency, typical for device connectivity and server synchronization; otherwise, system responsiveness and synchronization may be affected.

2.4.2.2.Available and Uninterrupted Power

LumineSense and its prototype assumes the presence of a stable Alternating Current (AC) power source in the classrooms where the system is located. As stipulated, the system relies on continuous power availability to operate sensors, microcontrollers, relay modules, and various network components. Thus, it is important to understand that power is a vital resource and the testing phase should

align with that assumption as well. Prolonged power interruptions may lead to temporary system downtime, delayed automation responses, or loss of unsynchronized data potentially causing the IoT system to record gaps among moments of blackout or power malfunctions.

2.4.2.3.Classroom and Environmental Suitability

Classrooms wherein the LumineSense system can be implemented to have varying degrees of focus and targeting all instances for the sake of versatility potentially harms the time-bound aspect of the project, thus, standardizing on the notion that classrooms are assumed to have followed the criteria of classroom dimensions is important to be considered. Additionally, the suitability for accurate sensor and data readings with respect to the components used are assumed to have minimal obstructions within the environment, both externally and internally.

2.4.2.4.Capabilities of Arduino Components

As mentioned, the embedded system comprises various components that the system must rely on, particularly on the Arduino Mega 2560 MCU, relay modules, and various sensors. These components are crucial to the deployment of LumineSense, and because of its potentially limiting constraints, such as processing speed, memory capacity, and availability of Input/Output (I/O) pins, these components may limit the number of sensors and devices that can be connected at once. Therefore, for deployment purposes, the system is intended for small to medium-sized classrooms due to these limitations. Furthermore, to prove that

usability, the project aims to ideate a system primarily focusing on interconnecting IoT towards the bulb mechanisms for the purpose of the implementation phase.

2.4.2.5.Shelf Life or Longevity of Components

Given that the LumineSense system uses components that inclines on accuracy, another factor to circumvent around is its longevity, which refers to the entire duration of the component until it lasts or does not work. This is prominent among the lighting mechanisms to be used in the prototype as its longevity or shelf life is important factor to consider when running the supposed IoT system in an actual classroom setting, which in turn requires even more maintenance in comparison to actual lighting procedures. Thus, the system does not focus on this matter because if interpreted, automation will not be is form but rather an input-based reliance.

2.4.2. Dependencies

Aside from the constraints and assumptions the LumineSense system must abide on, certain interdependencies give the system a foundational criterion to conceptualize on. These considerations influence the conception of the application's core functionalities as well as ascertaining that the project objectives are administered productively. LumineSense, with both hardware and software integrations, are identified to have the following dependencies that are critical for reduction and risk mitigation, holistically making sure that the project meets the needs of the users and the environment.

2.4.2.1.Functioning Components

The system integrates sensor readings and logical controls with displays so that accurate data may be presented. With this, numerous key hardware components must work properly for LumineSense to function with components such as the Arduino MCU, PIR, and the ESP32 Wi-Fi. Any deterioration in these parts interferes with features important to the system. Automation control, general operation, and data transfers as the general gist. Throughout the implementation phase, the dependence of the components is highlighted by the possibility of erroneous readings or complete system failure due to faulty connections or malfunctioning modules.

2.4.2.2.Active and Secure Server

A fully configured server needs to always be running, especially since LumineSense requires scheduling, energy reporting, and data storage management. Since web servers act as a bridge between the system's dashboard and IoT hardware, any malfunction or server downtime will disrupt system communication and functionality. The database, which contains the logging and analytical data, may also be subject to reliance because data loss and unauthorized access are among other concerns. In order to prevent these vulnerabilities, it is expected that the organization must keep the server running and functioning through routine backups and system updates. Moreover, with regards to security concerns, system administrators aside from the users of the LumineSense system are also important facets to consider aside from the project's physical assets to maintain.

2.4.2.3.Account Verification and Validation

Since the system also incorporates the management of a classroom's lighting, the importance of user controls and the level of privileged access that contrasts between each role is highlighted. Registration enables users of the LumineSense system to create accounts which grants them access to the general application functionalities. Validation of these created accounts becomes a concern when delineating identity concerns and authentication needs. Thus, the presence of an officiating body, particularly in an educational setting, that authenticates and validates accounts made by Faculty Members and Administrators is imperative.

This chapter provided an in-depth discussion of the technical framework and system features that define LumineSense. It explained how the system integrates IoT components, automation, and monitoring tools to achieve efficient classroom lighting management. Each subsection discussed the system's major functions, such as motion detection, gesture control, scheduling, and energy reporting–along with the necessary assumptions and dependencies for reliable operation. This chapter highlights LumineSense as a smart and efficient solution that helps schools save energy and improve classroom management.

CHAPTER III

SYSTEM FEATURES

This chapter outlines the key features of LumineSense designed to optimize energy efficiency and user interaction. Each module and functionality supports the system's primary goals of automatic lighting control based on occupancy and ambient light, augmented by intuitive user overrides. The system adjusts lighting levels based on occupancy detection, while providing flexible manual control through recognized gestures. Activity flow diagrams and functional requirements illustrate the processes, highlighting key user actions such as entering and exiting rooms, performing gestures for lighting overrides, and scheduling lighting behavior. Other corresponding system responses are discussed in the following sections.

3.1. System Decomposition

LumineSense has various components to consider, and this is scrutinized in the system decomposition. Each module in the breakdown serves a specific process to create an efficient, adaptive smart classroom lighting system. Overall, the decomposition encapsulates the project into a definite input, its appropriate processing, and the various output it delivers as described in the following sections.

As a project that capitalizes the use of embedded systems, LumineSense expects information from sensors and external devices, thus, such input data must be accurate. In finer details, the system receives data from two devices, namely, the PIR and the webcam. First, the role

of the PIR sensor is to serve as the module for detecting occupancy inside a classroom, which will activate the room's bulb for a set duration. The webcam grants the system an added convenience feature, detecting set gestures that act as an activation for various lighting mechanisms.

With proper input being fed into the IoT receivers, data that flows into it is ready for manipulation and transformation. Specifically for LumineSense, the information gathered from the sensors has roles for processing and control in line with lighting management. For information processing, first and foremost, the project relies on the microcontroller Arduino Mega 2560 to receive data then produce information. All hardware components are connected to this MCU which are then expected to generate appropriate outputs for data management across LumineSense. Additionally, with gesture controls as an added feature, appropriate software with built-in mechanisms that enables the user to capitalize on convenience and quality of experience by transforming gestures in place of manual lighting control.

Between these processing techniques there exists modules that actualize system functionalities as well as mitigating expected inaccuracy. By scrutinizing that notion, LumineSense relates to two main control mechanisms, namely for gesture recognition and logic shifters for the lighting grid. By considering the power output and input of various components, voltage regulators, the AMS1117-3.3V to be specific, enable safer implementation by controlling the light emission that the bulb allows, especially in the Wi-Fi connectivity. On the logical level of the system, shifters and relay modules allow further safer control by enabling light mechanisms in terms of turning different zones on or off with addition of logic level shifters of voltage control.

With various information going in and out of LumineSense, management over such objects is a vital concern to face. As detailed previously, the project requires a dependable and active server in order for the web dashboard to rely on and display data tables from. From these recorded data, the entirety of the IoT enables its various functionalities especially in the scheduling and reporting system. For schedule-based light management, timetables are of the essence wherein the sensor's detection is only allowable based on its time range which is where the reporting system comes into functionality: it manages instances of detection that goes out of defined timetable.

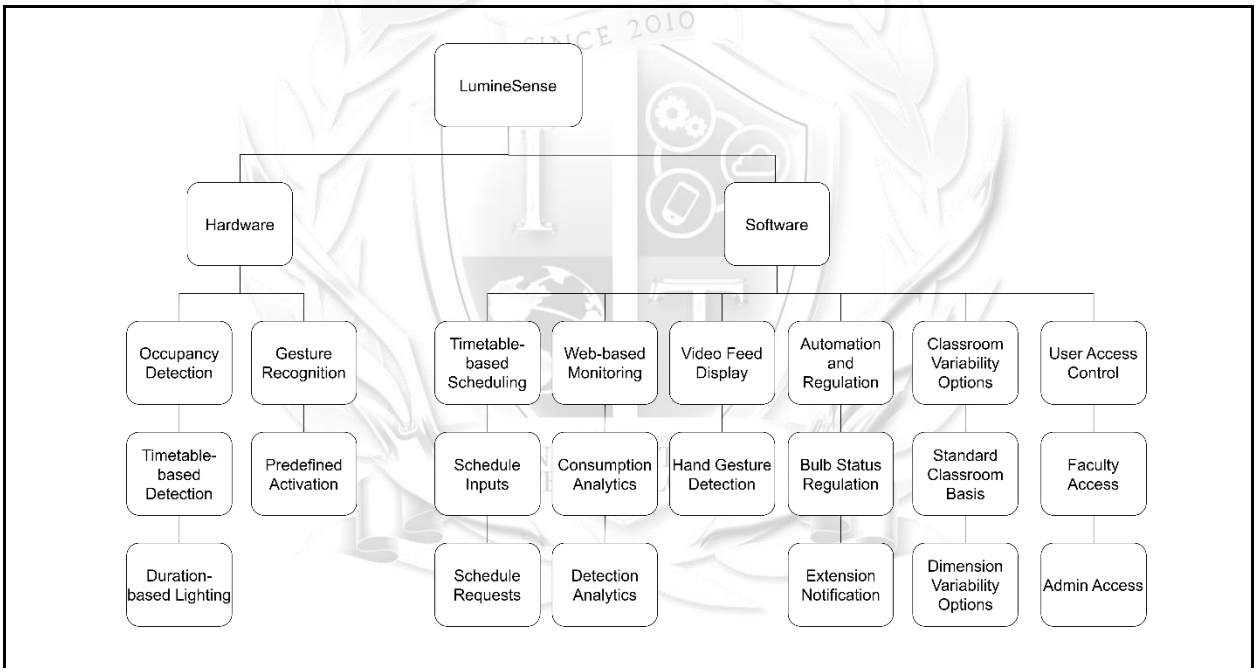


Figure 5. LumineSense Decomposition Chart.

3.2. System Functionalities

The functionalities for this project are centralized on it being an embedded system, capable of sourcing and detecting environmental information, and afterward, passing it onto software

platforms used throughout the application. Therefore, the main functionality of LumineSense is its capability to capture motion and heat, which will signal the bulb for automated lighting. From this capability, the additional functionalities of LumineSense branch out.

When the light bulb is signaled to turn on, its data is sent and stored in a server, from where systematic reports can be generated based on the energy conserved using the IoT system. Second, another integral feature is schedule-based management, wherein lighting is based on a specific timetable set according to class schedules. In this way, users can control when lighting is appropriate in a classroom, avoiding unnecessary energy usage on out-of-class hours. Finally, via wecamer, various gestural controls are integrated for seamless control over lighting, handling light management in swift motions.

Additionally, with the existing dashboard LumineSense has, the users can view the classroom where the bulb has been installed, the conditions of the lighting system, the number of hours elapsed when the sensors have detected movement, the specific timetable for a classroom, a brief description of the classroom, and a report analytics for energy consumed during operation, depending on the role chosen when registering. This enables the users to have a quick rundown of each component the project has, all the while maintaining the performance and operational efficiency of the IoT system.

In hindsight, hardware and software functionalities are detailed in the following sections with respective activity diagrams representing the flow of data until such function completes and how different actors receive and retransmit it. The acknowledgment of these specifics enables the continuity of data reception and transmission in the LumineSense system.

3.2.1. Occupancy Detection

One of the main aspects of LumineSense is the external devices it has embedded such as the PIR sensor. In essence, the system has the environment as its focus because without proper detection, the functionalities LumineSense has would be lacking and inevitably be useless. In terms of the motion detected by the PIR sensor, once it has sensed changes, it signifies that an occupant has entered the facility or room which would turn the bulbs into the specified vicinity. Therefore, the concept of having properly functioning hardware components is vital and in turn, a vital requirement for detecting occupancies to activate lighting mechanisms.

3.2.1.1.Descriptions and Priorities

As a lighting control and management system, importance of implementing contactless activation in contrast to manual lighting is vital. In LumineSense, this capacity is highlighted by sensor-based capabilities. In occupancy detection, the PIR sensor plays a major role in sending signals to turn lighting on or off based on environmental data. Aside from this, the priority of this module is entirely based on the occupant's behaviors because the data that the PIR sensor receives is dependent on the occupant's presence based on their motion and the heat emitted. Once a person enters the room with the system on and the sensor detects their presence, lighting mechanisms set in the program activate.

3.2.1.2.Stimulus/Response Requests

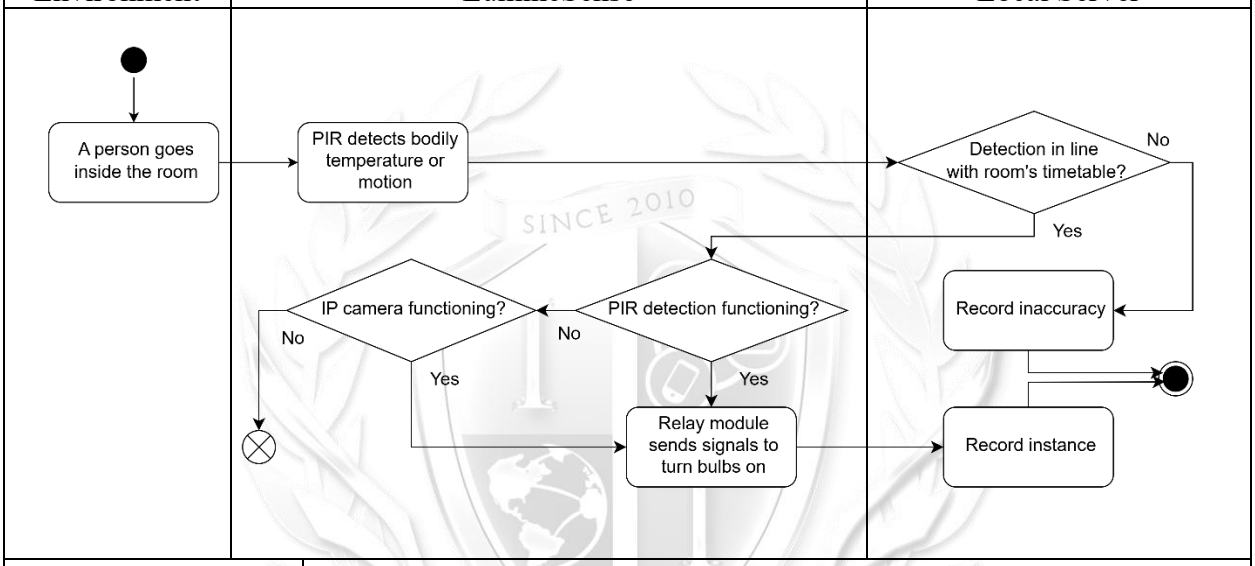
The PIR sensor is central to the system's capability of detecting occupancies. As shown in Table 13, whenever the sensor detects bodily temperature or motion, given its sensor accuracy and range, the system then relies on the classroom's defined timetable. This means that without the timetable or schedule for the sensor to base upon, it would be inefficient to regard these sensors when compared to manual lighting in the first place. Therefore, the timetable for schedule-based management is important here, which can be managed throughout the system's dashboard.

Regardless, if the classroom has a timetable defined, the light bulbs installed across the ceilings will turn on for a set duration, typically during in-class hours or when the sensor does not detect information for a set amount of time. Otherwise, the system will have to notify the user that the sensor has detected a discrepancy, meaning it detects outside of the scope of the timetable, if it has one. In terms of data management, all information is recorded specific to the time where the bulb has turned on or off, whether it was accurate or not based on the classroom's defined timetable.

If an instance wherein the PIR sensor cannot detect environmental or occupancy data due to various reasons such as possible malfunction and disconnection, the user will be informed in the web-based dashboard to calibrate the system setup and connections, to which certain alternatives and mitigations are also mentioned previously tailored towards user cautiousness.

Function Name	Occupancy Detection
Function Description	The PIR sensors, located close to the MCU, detects motion and heat emitted from people who pass through it.
Precondition	○ The PIR sensor must be properly functioning. ○ The room has a definite timetable based on scheduling. ○ The local server is functioning.
Post Condition	○ The bulb is turned on for a set duration. ○ Energy consumed within the total duration the bulb has been turned on is recorded.
Activity Flow	

Environment	LumineSense	Local Server



Alternate Flow	○ If sensor detection is not successful, IP camera overrides serve as backup for detection. ○ The user will have to set the classroom's timetable.
Exception Flow	○ The system will also notify the users through the dashboard, namely if the PIR sensor is not detecting and it has detected motion or body heat outside the classroom's given timetable. ○ If the PIR sensor has not detected bodily heat or motion for a specific amount of time, the light bulbs will turn off.

Table 13. Occupancy Detection Activity Diagram.

3.2.1.3.Functional Requirements

The PIR sensor requires various checks to determine whether it is fully functional or not. This component is required for further system functionalities,

particularly in emitting signals from the relay modules for the bulbs to light on. Therefore, the developers have determined the following requirements to ensure that the PIR sensor captures data and sends appropriate signals efficiently.

a. Ensure that the detection range of the PIR sensor occupies most of the room so that the environmental data and its response is appropriate.
b. In instances where the PIR sensor does not work and the IP camera is used as an alternative, ensure that the camera's feed is functioning correctly by validating its connection from the website dashboard.
c. Avoid placing components in an obstructed view to allow efficient motion and heat sensing.
d. Ensure other components are properly connected and are functioning.

Bachelor of Science in
INFORMATION
TECHNOLOGY

3.2.2. Gesture Recognition

The gesture recognition feature of LumineSense enables users to control classroom lighting through predefined hand movements detected by the web camera. This function enhances user convenience by allowing contactless interaction, which is especially beneficial in classroom settings to enhance efficiency over control. If the gesture is recognized, the system automatically updates the lighting and logs the interaction through the local server, ensuring continuous usability. Additionally, to control lighting based on

the gesture performed, the reliance over relay switches is also vital since depending on the constraint set, the appropriate signals would have to be sent out to the bulbs.

3.2.2.1.Descriptions and Priorities

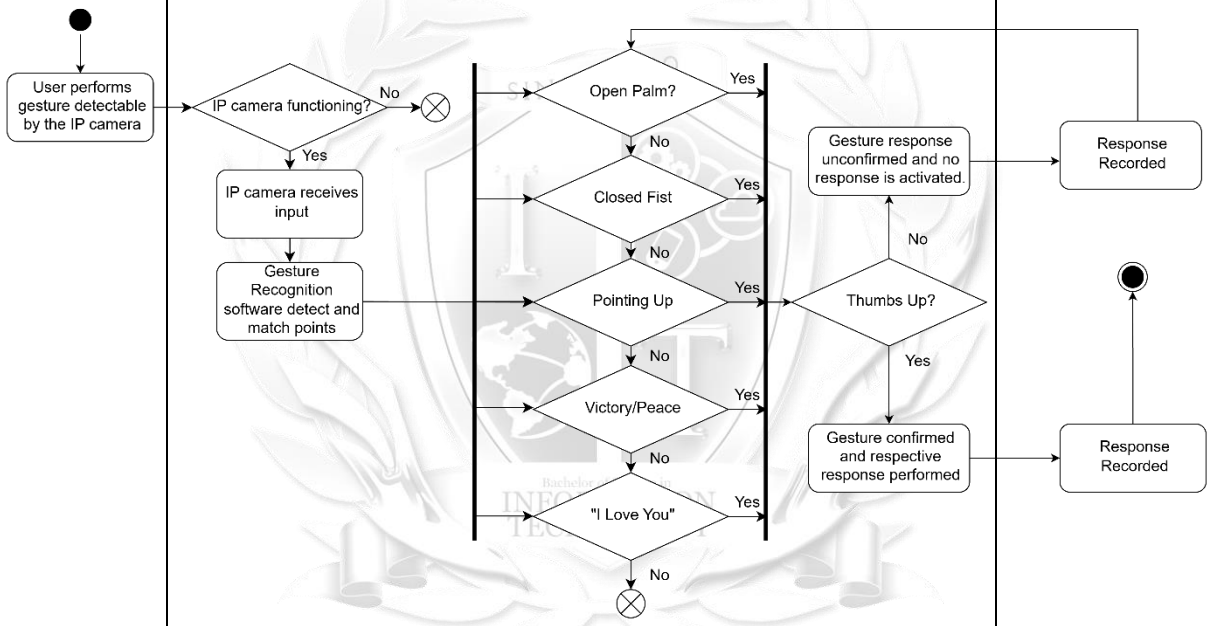
The web camera and the relay modules are the components required for this feature since the gestures performed by the user will be detected from the image sensing capability present in the camera. Gesture recognition software allows the system to detect the points of the user's digits and/or fingers, which in turn enables the relay modules to accept signals and pass it to the bulb for the appropriate response. Therefore, the priorities rely on the user's gestures and detecting capability of the recognition software as well as the functioning of other components such as the IP camera and the relay switches.

3.2.2.2.Stimulus/Response Requests

The predefined gestures the developers had set are the only input the gesture recognition software accepts. In turn, appropriate responses as shown in Table 14 are passed through the bulbs made possible by the relay switches. Requests of this feature are centered around the functioning of the components primarily around the capability of the web camera. If the camera is not functioning or its detection is not accurate, the user can use the web dashboard to alternatively do so. Any instance of gesture controls that the user made is also recorded in the local server, which is required for report and data analytics in other features.

Function Name	Gesture Recognition
Function Description	The web camera detects defined hand gestures for control and convenience.
Precondition	○ The web camera and gesture recognition software must be properly functioning. ○ Dimmer modules and relay switches are connected. ○ The local server is functioning.
Post Condition	○ Bulbs automatically adjusted based on the performed gesture. ○ Data recorded in the database.

Activity Flow

User	LumineSense	Local Server
		

Alternate Flow	○ Mistaken or accidental gesture-based activation is constrained behind a confirmation gesture. ○ If the gesture performed is inaccurate or does not exist in the defined set, the user will alternatively use buttons in the dashboard that serve the same purposes. ○ Timed web-based popups also serve as fallbacks for when the confirmation gesture is performed.
Exception Flow	○ Through the dashboard, the system notifies the user if the IP camera is not functioning.

Table 14. Gesture Recognition Activity Diagram.

3.2.2.3.Functional Requirements

The main components involved in the gesture recognition feature are the web camera and the relay modules. These components enable the system to detect hand movements and interpret them into lighting control commands sent as signals by the relay switches. Thus, considering the visual nature of gesture detection and the electrical mechanism involved in the processing, accuracy and safety should be focused on alongside the feature's functional requirements.

e. Ensure that the web camera has an unobstructed view where the user will perform gestures.
f. Confirm that the OpenCV and MediaPipe libraries are properly installed and configured.
g. Ensure stable connection between the IP camera and the MCU for real-time gesture processing.
h. Ensure all other components are properly functioning to support gesture override capabilities.
i. At instances where the IP camera receives inaccurate input, alternative dashboard controls can be used by the user.

3.2.3. Timetable-Based Scheduling

The timetable-based scheduling feature enables administrators to define and control classroom lighting operations with predefined timetables that align with operational hours and academic schedules. This functionality provides an approach in energy conservation

by enabling sensor-based lighting only operable during designated time period. In turn, this prevents unnecessary power consumption during non-instructional hours. Through the integration of the DS3231 RTC module and the web-based dashboard, accurate timekeeping is maintained.

The timetable-based scheduling feature enables administrators to define and control classroom lighting operations with predefined timetables that align with operational hours and academic schedules. This functionality provides an approach in energy conservation by enabling sensor-based lighting only operable during designated time period. In turn, this prevents unnecessary power consumption during non-instructional hours. Through the integration of the DS3231 RTC module and the web-based dashboard, accurate timekeeping is maintained, which allows authorized personnel to create, monitor, and modify lighting schedules tailored to each room's unique lighting patterns.

3.2.4.1.Descriptions and Priorities

This feature requires the DS3231 RTC module to maintain accurate time tracking, the local server to store timetable data, and the web-based dashboard to provide interface for management and schedule creation. Moreover, the scheduling system is expected to integrate seamlessly with the PIR sensors, which actively control lighting. This module's priority depends on the accuracy of the RTC module and authorized users setting up timetables correctly because improper scheduling might lead to inadequate illumination during actual class sessions.

A hierarchical control feature is the basis of this scheduling feature since defined timetables takes precedence over default sensor behaviors. When a classroom timetable indicates non-operational hours, the system disables automatic lighting regardless of motion detected which is conversely recorded in the local server whenever such an instance occurs. However, during scheduled in-class hours, the sensor-based automation functions as usual, providing responsive lighting that adapts to occupancy conditions.

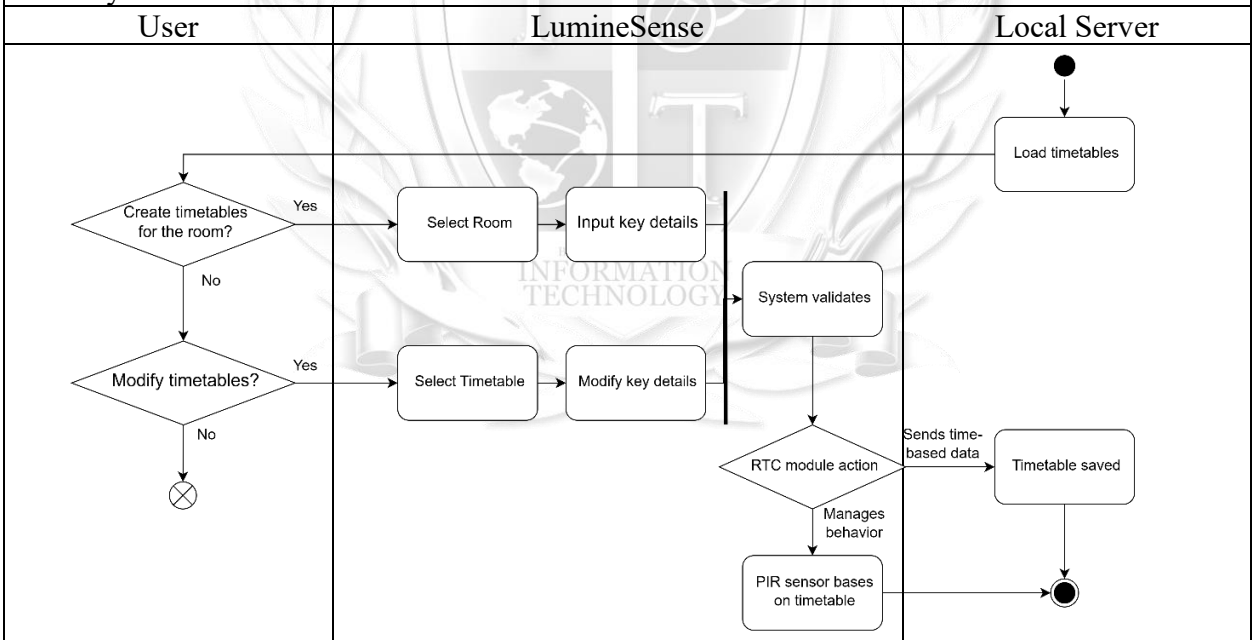
3.2.4.2.Stimulus/Response Requests

Administrators configure weekly timetables through the web-based dashboard, defining operational hours for each classroom. As shown in Table 15, the system validates and stores schedule inputs in the local database, then synchronizes them with the corresponding MCU. The DS3231 RTC Module supplies continuous time data to the Arduino Mega 2560, which evaluates whether sensor-based automation should operate based on the defined schedule.

Within scheduled hours, the system activates full sensor functionality by which the PIR sensors detect occupancy. All lighting events during these periods are recorded as standard operations. Beyond scheduled hours, the system transitions to restricted mode, monitoring sensor inputs without triggering lighting responses. Motion detected by PIR sensors or gestures identified by web cameras outside operational windows are logged as security anomalies, prompting dashboard notifications to alert administrators of potential unauthorized presence. The system supports manual override

capabilities for exceptional circumstances. Authorized users can temporarily extend operational hours via the dashboard to accommodate special events, extended sessions, or maintenance requirements.

Function Name	Timetable-based Scheduling
Function Description	Users set timetables to allow sensor-based lighting automation during operational hours and otherwise prohibit activation during non-operational hours.
Precondition	○ The user initializes the LumineSense system through the web-based dashboard. ○ The DS3231 RTC module is functional. ○ The local server is functioning.
Post Condition	○ Operational hours for the room's sensor-based automation are set. ○ Data is recorded in the database.
Activity Flow	



Alternate Flow	○ Administrators can extend operational hours beyond usual class hours.
Exception Flow	○ If in cases the system doesn't detect timetable data, the system is based on schedule saved from the SD card which in turn, would notify the user.

Table 15. Timetable-Based Scheduling Activity Diagram.

3.2.4.3.Functional Requirements

The DS3231 module for precise timekeeping, the local database for timetables and schedule-related logs, and the web-based dashboard for user interaction are the primary parts of this schedule-based management feature. Both data integrity and time accuracy should be taken into consideration in addition to the functional requirements that the developers have specified for the component because of the crucial nature of time-dependent operations and the possibility of schedule conflicts.

a. Verify that the RTC module maintains accurate time with minimal drift caused by external factors such as temperature changes around the system.
b. Ensure that input validation prevents overlapping schedules or invalid time modifications.
c. Confirm that updates schedules are transmitted from the dashboard to the Arduino MCU without delay.
d. Validate that sensor detection outside scheduled hours is properly logged as security events rather than operational data.
e. Notify administrators when RTC issues are detected.

3.2.4. Web-based Monitoring

By combining real-time sensor data, lighting status, energy metrics, and system health information from all monitored classrooms, the web-based monitoring dashboard

gives authorized staff centralized access to LumineSense functionalities. The dashboard, which was created with HTML, CSS, JavaScript, and backend technologies integrated with XAMPP, converts unprocessed IoT data into user-friendly controls and visualizations that allow for well-informed decision-making without the need for technical knowledge.

System reactions are triggered by user interactions with the dashboard. When an administrator makes changes to a classroom schedule, the dashboard verifies the information, updates the database, and uses the ESP32 Wi-Fi module to send the updated schedule to the appropriate MCU. When a user requests an energy consumption report for a particular time period, the dashboard retrieves historical data from the database, computes the energy savings over baseline manual lighting, and produces reports that can be downloaded in easily readable formats. When dashboard buttons or sliders are used to initiate manual lighting control commands, the corresponding relay modules receive the commands instantly and provide responsive feedback that verifies the action was carried out.

Bachelor of Science in INFORMATION TECHNOLOGY

3.2.5.1.Descriptions and Priorities

The requirements for this feature are stable local network connectivity, a running XAMPP server, and constant data transfer from ESP32 Wi-Fi modules attached to the classroom's Arduino MCU. The dashboard needs to synchronize with the MySQL database that holds schedules, consumption metrics, sensor states, and historical logs. The priority of this feature depends on server uptime and network infrastructure dependability because interruptions momentarily impair monitoring capabilities.

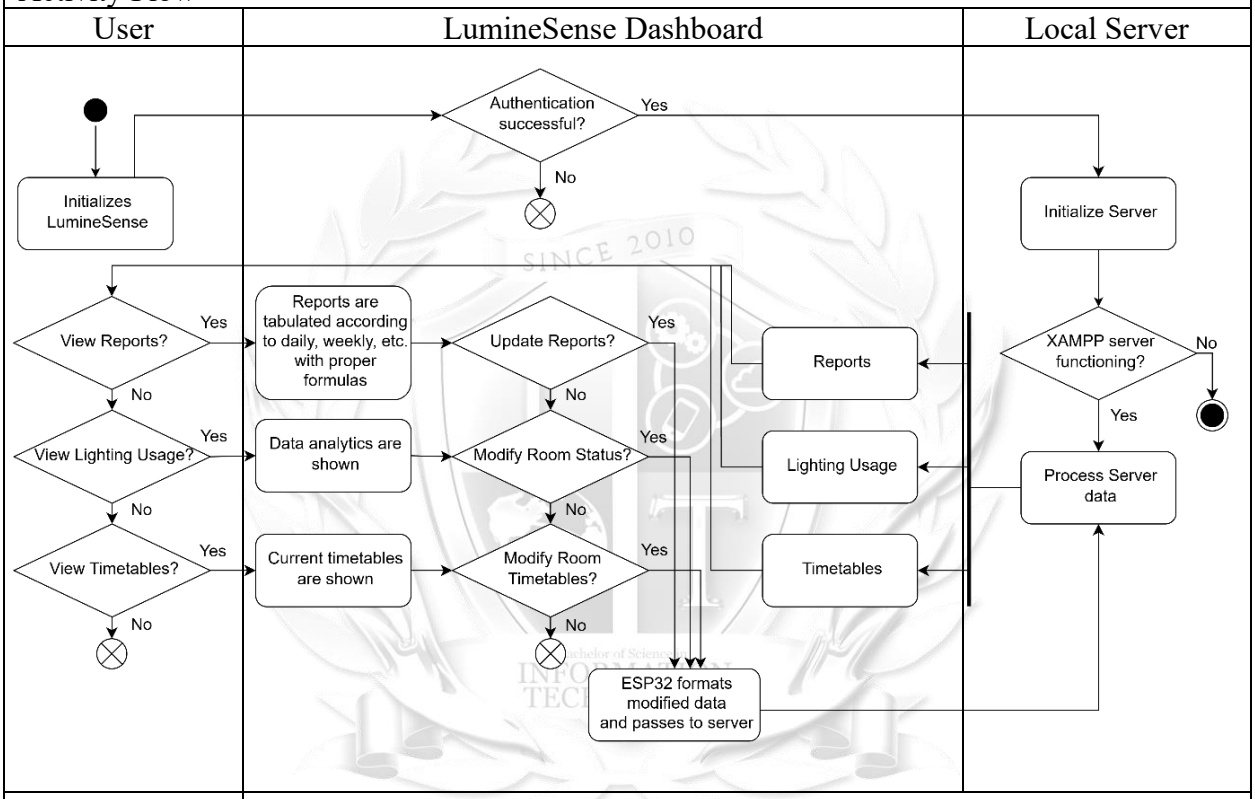
A hierarchical control feature is the basis of this scheduling feature since defined timetables takes precedence over default sensor behaviors. When a classroom timetable indicates non-operational hours, the system disables automatic lighting regardless of motion detected which is conversely recorded in the local server whenever such an instance occurs. However, during scheduled in-class hours, the sensor-based automation functions as usual, providing responsive lighting that adapts to occupancy conditions.

3.2.5.2.Simulus/Response Requests

The dashboard loads the relevant interfaces according to permission levels after verifying and authenticating user credentials. Real-time updates take place through periodic data refreshes without the need for manual page reloads, as Table 17 illustrates. Unusual consumption patterns, motion during non-scheduled hours, or component malfunctions are examples of significant events that cause visual alerts through color-coded notifications that remain visible until acknowledged.

The system reacts instantly to user interactions. Modifications to the schedule are verified, saved in the database, and sent via ESP32 modules to the appropriate MCUs. Energy report requests compute savings over baseline manual lighting calculated by its appropriate formulas, query historical data, and produce reports can also be downloaded. When requested, the dashboard shows IP camera feeds with real-time gesture recognition overlays that validate recognized patterns and detected hand positions.

Function Name	Web-based Monitoring
Function Description	This feature allows users to view essential data from the LumineSense system through a web dashboard, primarily
Precondition	○ XAMPP server is running and accessible. ○ Local MySQL database is functioning and contains system data. ○ ESP32 Wi-Fi is connected and transmits data from the MCU. ○ Users have validated their credentials.
Post Condition	○ Dashboard displays updated information reflecting current system status and performs executable actions.
Activity Flow	



Alternate Flow	○ If network connectivity is interrupted, the dashboard displays cached or local data from the Micro SD card.
Exception Flow	○ Users are encouraged to re-enter credentials with error messages indicating invalid username or password if authentication fails.

Table 16. Web-based Monitoring Activity Diagram.

3.2.5.3.Functional Requirements

The main components involved are the XAMPP server, MySQL database, the ESP32 Wi-Fi module, and user web browsers. Due to the

dashboard's role as the primary human interface, both security and usability are critical alongside functional requirements.

a. Verify if the XAMPP server stability and implements data backup procedures to prevent data loss during maintenance.
b. Enfore strong authentication and role-based access controls.
c. Ensure that the dashboard displays updates real-time or every 5-10 seconds with accurate sensor data and lighting status that reflects actual bulb states.

3.2.5. Video Feed Display

By employing a webcam set up for gesture recognition, the live camera feed display offers real-time visual access. The system provides contextual visual information to supplement sensor-based data by allowing video feed through the web dashboard. This makes gesture controls specifically for the faculty members or instructors using the LumineSense system to enable automation of gesture controls.

3.2.6.1.Descriptions and Priorities

The necessary functionalities for the webcam are its link to the local network, ESP32 Wi-Fi module, and enough bandwidth for video transmission without delaying latency. When monitoring multiple classrooms, the dashboard’s video rendering components support multiple simultaneous feeds. The feature’s priority is based on camera performance and network stability, since quality affects the service for gesture recognition.

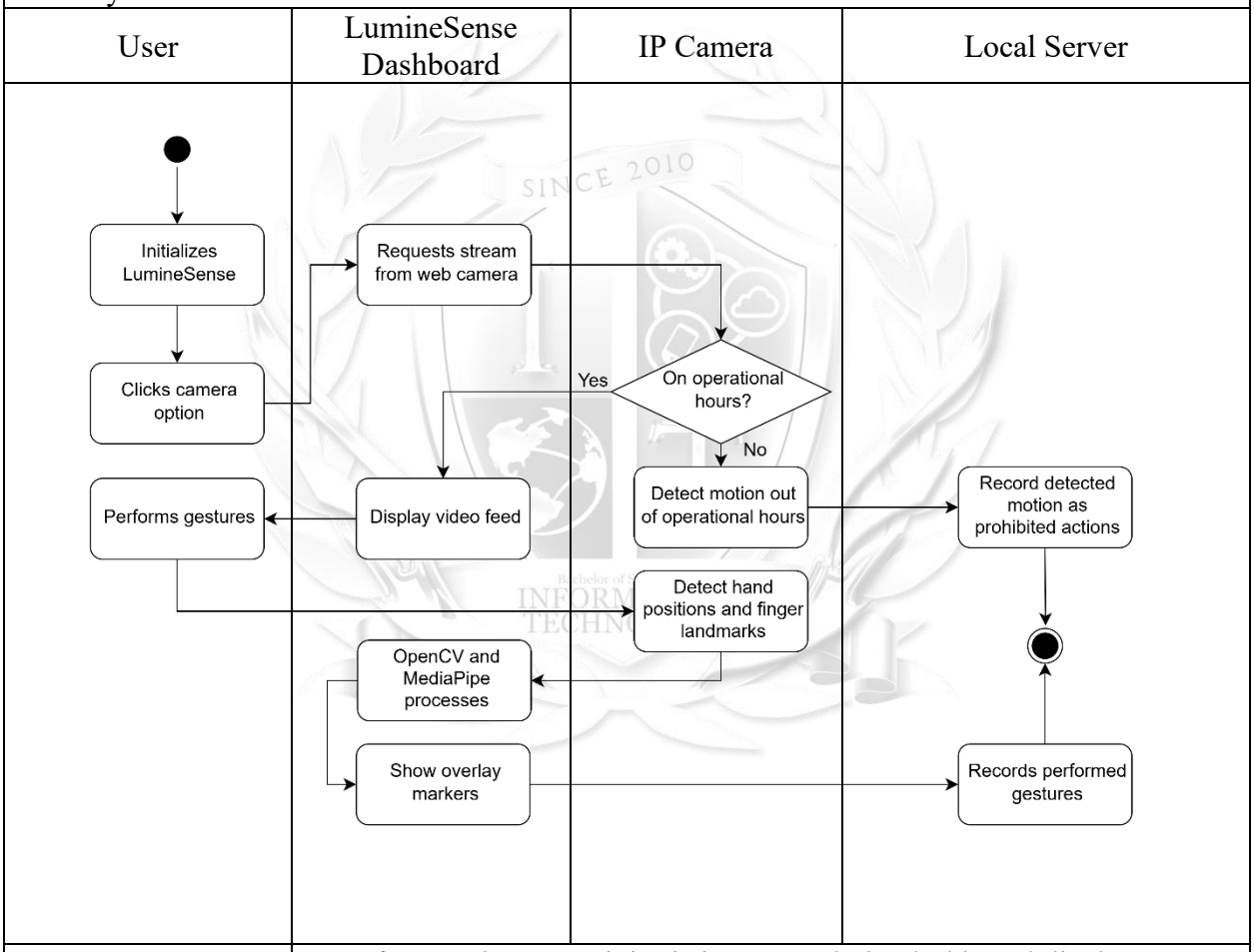
When a classroom timetable indicates non-operational hours, the system disables automatic lighting regardless of motion detected which is conversely recorded in the local server whenever such an instance occurs. However, during scheduled in-class hours, the sensor-based automation functions as usual, providing responsive lighting that adapts to occupancy conditions.

3.2.6.2.Stimulus/Response Requests

As shown in Table 18, after initializing the application, when users view classroom camera via icons or security alerts, the system establishes web camera connections and streams video to the browser. By default, the webcam feed is turned off to avoid access disruptions among non-users. To enable such functionality, the user has to A screen displaying the recorded feed of the webcam loads within dashboard panels alongside lighting status, sensor readings, and schedule details, continuing until users navigate away the home page or by explicitly closing the connection.

When users perform defined gestures, markers indicate detected fingertips and landmarks made possible through the gesture recognition software. As the system recognizes valid gestures, overlays that notify users briefly highlight matched patterns before executing lighting commands, providing immediate confirmation of successful recognition. Upon recording such gestures, whether correct or incorrect, the system then records such instance for data relevance in report management in the administrative side of the application.

Function Name	Video Feed Display
Function Description	The webcam provides real-time video feed to the dashboard when enabled, which allows visual verification of the gesture recognition feature.
Precondition	○ The IP camera must be powered on and connected to the local network. ○ The ESP32 Wi-Fi module must be functioning. ○ The XAMPP server supports video streaming capabilities.
Post Condition	○ Live video feed is displayed within the web-based dashboard. ○ Gesture recognition overlays appear when hand movements are detected.
Activity Flow	



Alternate Flow	○ If network connectivity is interrupted, the dashboard displays cached or local data from the Micro SD card.
Exception Flow	○ Users are encouraged to re-enter credentials with error messages indicating invalid username or password if authentication fails.

Table 17. Video Feed Display Activity Diagram.

3.2.6.3.Functional Requirements

The primary parts are network-connected web cameras, XAMPP streaming infrastructure, native camera networking or ESP32 modules, and browser video rendering capabilities. Alongside functional requirements, privacy concerns and performance dependability are crucial. The developers have determined the following requirements to ensure full functionality.

a. Verify if the web camera maintains consistent network connection.
b. Verify if the web camera responds well with streaming requests by checking device permissions.
c. Configure bitrate streaming by adjusting quality based on bandwidth, ensuring feeds load within seconds and minimal latency persists.
d. Confirm MediaPipe hand tracking overlays render correctly without degrading the IP camera's performance.
e. To further security, log access events and alerts with user identification and timestamps.

3.2.6. Automation and Regulation

This feature relies on coordinated inputs from multiple subsystems to enforce intelligent automation rules while regulating unintended behaviors and false positives. The Arduino Mega 2560 executes control logic that evaluates occupancy detection, lighting schedules, electrical parameters, and user commands in real-time, making autonomous

decisions to optimize classroom lighting. Priority is given to safety regulations of the electrical components used, especially in the lighting mechanisms, and overall operational efficiency with respect to the IoT components.

3.2.6.1.Descriptions and Priorities

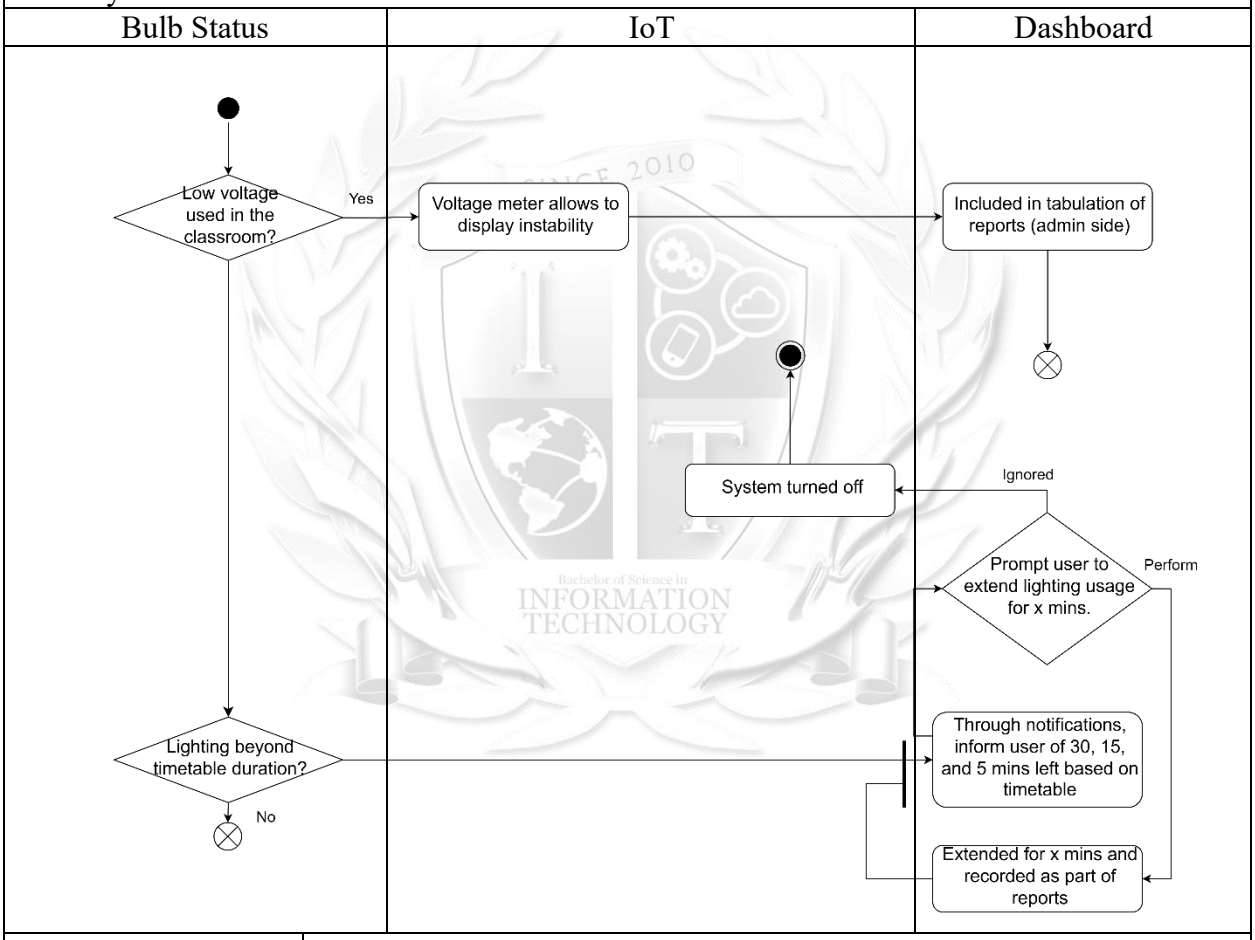
The main utilization of this feature relies on scenarios of the low voltage of lighting in the specific room which is where the priority of this feature resides. Specific instances such as bulbs flickering and no lighting often signals voltage instability, loose connections, or nearing failure in LED bulbs. In addition, underpowered bulbs that fail to meet lumen requirements for classrooms. Extensions of lighting usage are also focused on this feature wherein faculty members can have the option to extend room usage outside of the prescribed timetable of the system.

3.2.6.2.Stimulus/Response Requests

The response of the system regarding bulb-related deficiencies enables the relay modules to control and isolate faulty bulbs to prevent hazards, while logging the event for maintenance reports. A digital voltage meter and CT sensors exist to detect the current flow of all the bulbs implemented in the design while software-based mechanisms allow the users to confirm action being done. With situations forementioned, the system continuously monitors current and voltage via sensors connected to the Arduino MCU to which exclusive alerts are sent to the dashboard in the form of popups. In the administrative side, these instances are recorded for the maintenance purposes.

Function Name	Automation and Regulation
Function Description	The system is capable of handling additional features in line with automation of bulbs and system settings as well as regulations of restricted and overlapping use.
Precondition	○ The bulbs are successfully installed and connected to the system. ○ Voltage multimeters, CT Sensors, ESP32 Wi-Fi, and RTC are functioning. ○ Dashboard and local server is functioning.
Post Condition	○ Unusual bulb activities are notified in the dashboard. ○ Exclusive warnings and prompts are displayed depending on level of importance.

Activity Flow



Alternate Flow	○ When no voltages or wattages are used, the system sends a notification that potentially, there could be no bulbs installed or there are problems in connections.
Exception Flow	○ If the ESP32 Wi-Fi did not receive any data, the user is required to validate Internet connectivity issues.

Table 18. Automation and Regulation Activity Diagram

3.2.6.3.Functional Requirements

The automation and regulation feature requires robust logic processing, accurate sensor inputs, and fail-safe mechanisms to function reliably. The following requirements ensure this feature operates as intended:

a. Ensure the Arduino Mega 2560 executes control logic with minimal latency
b. Calibrate and ensure that the Voltage Multimeters and CT Sensors are functioning and are connected to the bulbs being used.
c. Ensure the bulbs being used are connected properly.
d. Ensure that the predefined timetable is in line for the room to ensure lighting extension beyond the schedule is accurate.

3.2.7. Classroom Variability Options

The classroom variability options feature provides administrators with flexible configuration settings that adapt LumineSense to diverse classroom dimensions, occupancy patterns, and lighting requirements. By recognizing that educational institutions utilize spaces ranging from small tutorial rooms to large lecture halls, this feature allows the system to scale its hardware deployment and software logic to match physical environments. By offering predefined templates based on standard classroom sizes while also permitting custom configurations, LumineSense ensures optimal performance across varied institutional contexts.

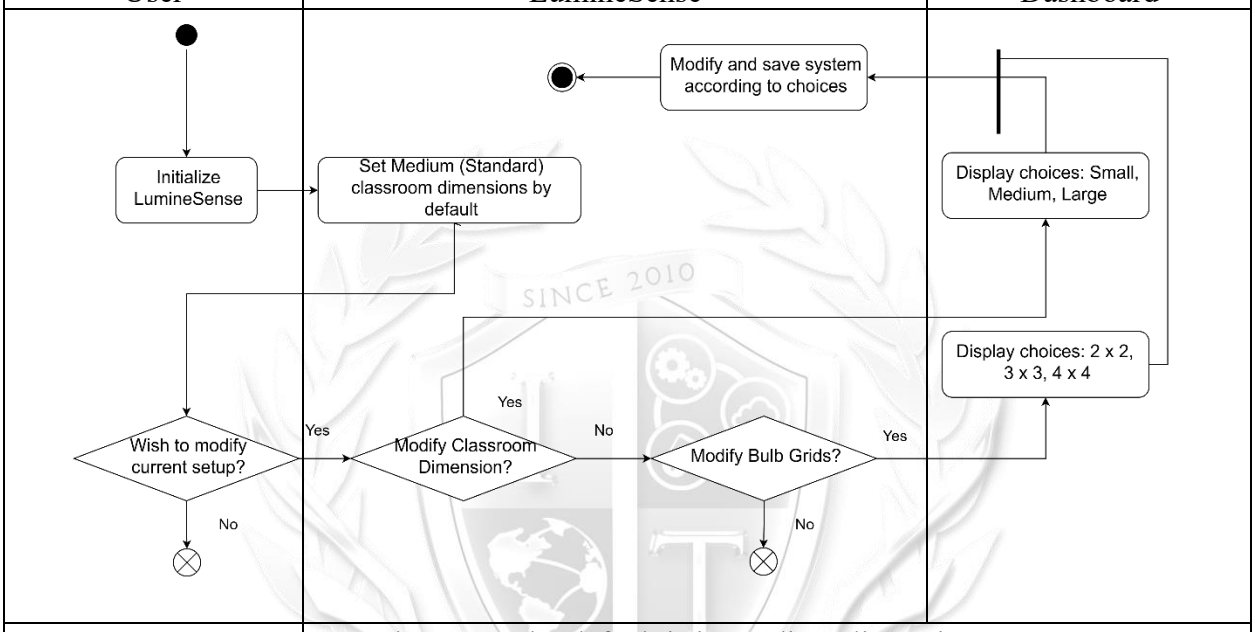
3.2.7.1.Descriptions and Priorities

This feature relies on the web-based dashboard interface to present configuration options during initial system setup or when expanding deployments to additional classrooms. The system defaults to medium classroom dimensions (7m x 9m, 63 sq. m.) with a 3x3 bulb grid, reflecting the prototype design and alignment with DepEd Order No. 64 standards. However, administrators can select small classroom configurations or large classroom configurations based on actual room dimensions.

3.2.7.2.Stimulus/Response Requests

When administrators access the classroom configuration module through the dashboard, the system presents three primary size templates alongside a custom configuration option. By selecting "Small," the system activates settings optimized for 49 sq. m. spaces. In the contrary, selecting "Medium" (the default) maintains nine bulb zones. Lastly, selecting "Large" configures relay logic for 8x8 bulb zones, ensuring uniform coverage in expansive lecture rooms. When variability options are modified after initial deployment, the system prompts administrators to update hardware components if necessary and recalibrates baseline energy consumption metrics used for savings calculations. All configuration changes are logged with timestamps and user identifiers to maintain an audit trail of system modifications. If the configuration is inadequate, the system displays warnings and recommends increasing the bulb count or adjusting room usage expectations.

Function Name	Classroom Variability Options
Function Description	By default, the system is based on a Medium dimension with 3 x 3 bulbs setting (7 m x 9 m, as prescribed in System Specifications) in which the system's prototype is based on. Additional options are presented and are available for purpose of variability.
Precondition	○ The LumineSense system has been successfully initialized.
Post Condition	○ Desired classroom or dimension modification is set.
Activity Flow	



Alternate Flow	○ The system by default is in Medium dimensions. ○ The user can change the number of lights that are otherwise set in other classroom dimensions in accordance by their preferences which is required to be confirmed by the Administrators.
Exception Flow	○ If the LumineSense system did not initialize properly, the user is required to validate Internet connectivity issues.

Table 19. Classroom Variability Activity Diagram.

3.2.7.3.Functional Requirements

The classroom variability options feature requires intuitive configuration interfaces, accurate calculation, and clear guidance for

administrators. The following requirements ensure this feature operates as intended:

a. Ensure dashboard configuration displays visual representation of the bulb grid layouts.
b. Communication with the system's administrator allows to the Faculty Member to externally validate the rooms dimensions and to further reassess with required specifications..

3.2.8. User Access Control

This feature establishes a hierarchical permission system that governs which individuals can view, modify, or interact with LumineSense functionalities. By recognizing that educational institutions have diverse positions and roles, Administrators are therefore responsible for system management and Faculty Members as general users for general system functionalities. This feature ensures appropriate access levels while maintaining operational security. By implementing role-based authentication and authorization mechanisms, LumineSense protects sensitive configurations from unauthorized changes while providing intuitive interfaces tailored to each user class.

3.2.8.1.Descriptions and Priorities

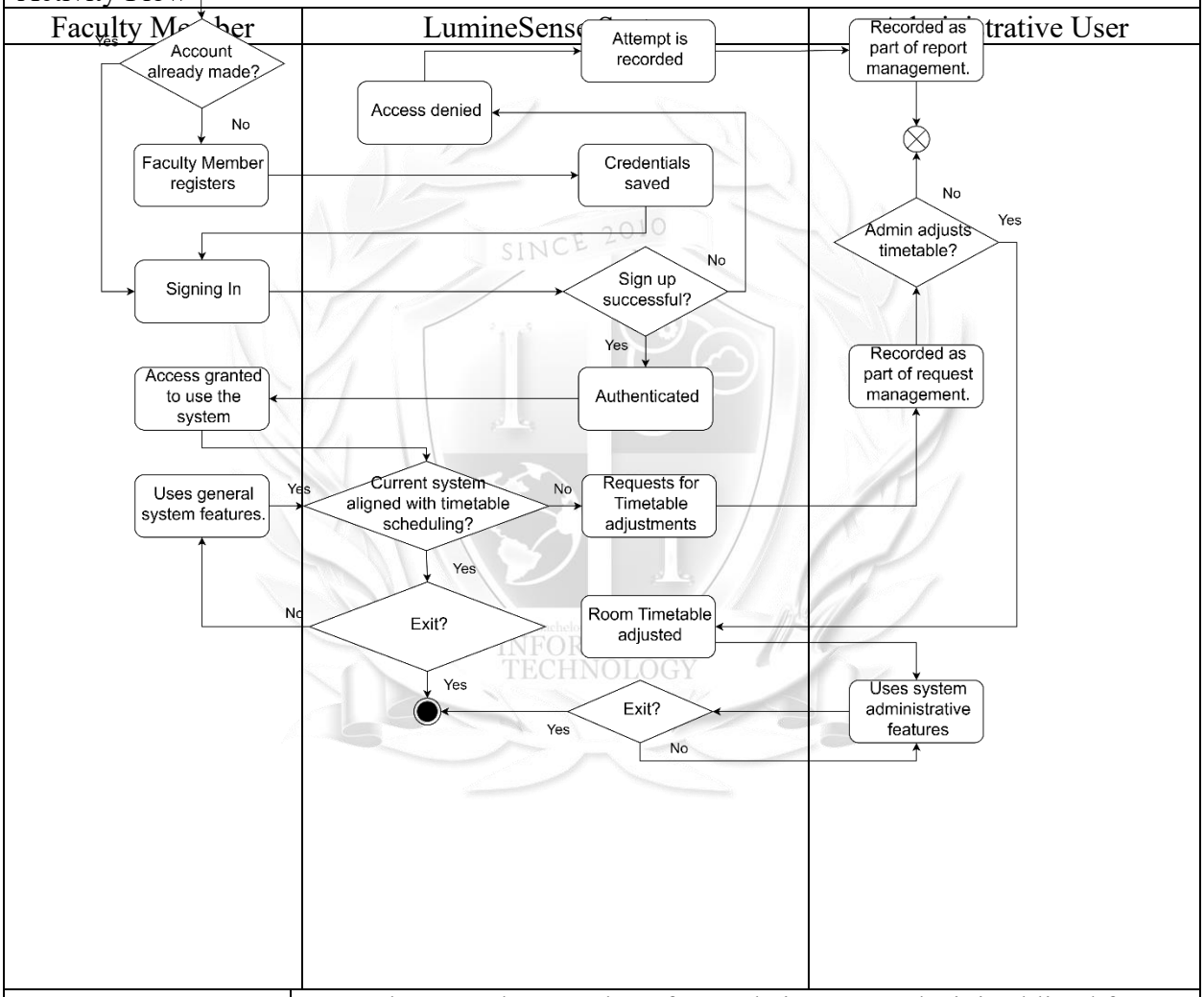
This feature relies on the local database to store user accounts, role assignments, and access logs, with authentication handled through the web-based dashboard login interface. The system defines two primary user roles:

Administrative Users possess full system access, including the ability to create schedules, modify classroom configurations, view energy analytics, manage user accounts, and adjust system settings. Faculty Members have limited access focused on viewing real-time classroom status, utilizing gesture controls during teaching sessions, and submitting lighting adjustment requests to administrators. Priority is given to preventing unauthorized access to administrative functions that could disrupt automated lighting operations, compromise energy conservation goals, or expose sensitive institutional data with respect to the Data Privacy Act of 2012.

3.2.8.2.Stimulus/Response Requests

When users access the LumineSense dashboard, the system presents a login interface requiring valid credentials. Upon receiving a username and password, the system queries the local database to verify account existence and password correctness. Administrators logging in receive full dashboard access, including schedule management interfaces, energy analytics displays, user account administration panels, and system configuration tools. On the other hand, Faculty Members logging in see simplified interfaces displaying real-time classroom status, occupancy indicators, and links to instructional materials explaining gesture controls. If authentication fails, the system displays generic error messages to prevent information leakage about account existence. After three consecutive failed login attempts within 15 minutes, the system implements a temporary lockout.

Function Name	User Access Control
Function Description	General users of the system and its overall functionalities are derived by the level of access and control between Faculty Members and Admins.
Precondition	○ Accounts for main users have been successfully registered and validated. ○ The local server storing user accounts is functioning.
Post Condition	○ Access for Faculty Members and Administrators is authenticated and can be used by the system according to their level of privilege access.
Activity Flow	



Alternate Flow	○ When Faculty Members forget their passwords, it is obliged for them to follow necessary procedures to change credentials in compliance with the Administrators.
Exception Flow	○ If the LumineSense system did not initialize properly, the user is required to validate Internet connectivity issues.

Table 20. User Access Control Activity Diagram.

3.2.8.3.Functional Requirements

The user access control feature requires secure authentication mechanisms, robust authorization logic, and comprehensive logging capabilities. The following requirements ensure this feature operates as intended:

a. Ensure passwords are encrypted in the local server for layered security.
b. Confirm that the system logs all authentication events are with timestamps and user identifiers.
c. Provide administrators with user management interfaces that display report handling, timetable request handling, active sessions, recent login history, and role assignments.
d. Validation and authentication of created Faculty Members accounts are handled by the Administrator Users.
e. Validation and authentication of created Administrator accounts are handled by the school's information security office.

This chapter highlights a comprehensive scrutiny of the core functionalities and modules that enables the LumineSense system to operate as an integrated IoT smart lighting management and monitoring system for educational institutions. This chapter began by presenting the system's decomposition, which outlined how processing units, input devices, control mechanisms, and data management infrastructure work cohesively to deliver automated and adaptive user-controlled lighting solutions.

CHAPTER VI

PROJECT MANAGEMENT

This chapter outlines the strategic project management procedures of LumineSense. It includes group decision-making in user identification, feasibility assessment, project timeline, role facilitation, and team communication. These factors have an impact on product development and deployment. In addition, this section outlines the key project management aspects, each of which contributes to the successful execution and implementation of the final product.

6.1. User Classes and Characteristics

The user classes represent the distinct groups of individuals who interact and utilize LumineSense. These users have different levels of technical expertise, responsibilities, and interaction frequencies with the system. The primary user classes include facility managers who oversee energy consumption and lighting operations. Each user class has specific characteristics that define their interaction patterns, technical proficiency, and system requirements within the classroom environment.

6.1.1. Facility Managers and Staff

Facility Managers and Maintenance Staff are the primary everyday users of LumineSense who oversee the operational aspects of classroom lighting and energy management. These users typically have moderate technical literacy and are responsible for monitoring for monitoring energy consumption patterns reviewing daily savings reports

and ensuring optimal lighting conditions across multiple classrooms. They utilize the web-based dashboard to track energy use, consumption logs, analyze lighting patterns, and identify opportunities for waste reduction. Facility managers interact with features such as the energy analytics display, automated lighting controls, and daylight sensor data to make informed decisions about campus resource allocation and sustainability initiatives.

6.1.2. Students

Students represent an indirect user class that experiences the benefits of LumineSense through improved learning environments. These users have no direct interaction with the system interface and possess minimal awareness of the underlying technology. Students benefit from automated lighting that adapts to natural daylight conditions, reducing eye strain and creating more comfortable study environments. While students do not actively engage with LumineSense, their presence triggers the motion sensors that drive the system's occupancy-based lighting conditions.

6.1.3. Faculty Members

Faculty Members are indirect users of LumineSense who benefit from the automated lighting environment created within their classrooms during teaching hours. These users have no direct interaction with the system interface but experience the effects of optimized lighting conditions during the in-class hours. These users may need to understand the basic system operations, such as the 10-minute motion detection delay, understanding how daylight sensors adjust bulb brightness, and the gestures required to

interact with the system. Their feedback on lighting comfort and conditions can inform facility managers about necessary adjustments to schedules or sensor sensitivity settings.

6.2. Product Feasibility and Assessment

This section overviews the comprehensive feasibility assessments, including technical, marketing, and economic evaluations. It is the critical phase of project management as this investigates and ensures that LumineSense is viable and reliable. This segment evaluates the aspects of implementing core functionalities of the system while assessing the needs of users and the operation of the product within educational institutions.

6.2.1. Technical Feasibility

Technical feasibility examines whether LumineSense can be successfully developed and implemented using available technologies, hardware components, and software frameworks. This assessment evaluates the system's core functionalities, the availability and reliability of necessary sensors and controllers, the integration requirements with existing institutional infrastructure, and the technical expertise required for deployment and maintenance.

6.2.1.1.Core Functionalities

The core functionalities of LumineSense are technically feasible through the integration of available IoT components and proven technologies. Motion detection is achievable using PIR sensors that detect human presence in classroom environments. Gesture control functionality

is implemented using gesture recognition software that detect hand movements for intuitive lighting control, enabling users to turn lights on or off and control specific rows through simple hand gestures without physical contact.

The Arduino MCU platforms offer sufficient processing capabilities to handle sensor data inputs from motion, ambient light, and gesture sensors, execute control logic for relay and dimmer modules, and communicate with the web-based dashboard through Wi-Fi modules. The 10-minute motion detection delay and automated on or off switching mechanisms are easily programmable within the microcontroller environment. Web dashboard development using standard frameworks such as HTML, CSS, JavaScript, and backend technologies like Node.js or Python with Flask ensures that real-time monitoring, energy consumption analytics, schedule management features, and gesture control configuration can be implemented effectively.

6.2.1.2.Data Availability and Integration

Data availability and integration capabilities are critical for LumineSense to function as a comprehensive lighting management system. The system requires continuous data streams from PIR sensors for occupancy detection, LDR sensors for ambient light levels, gesture recognition sensors for manual control inputs, and relay status indicators for current lighting states across monitored classrooms. This sensor data is

readily available through hardware components that communicate via standard protocols such as GPIO connections to the microcontroller. Gesture sensors can be integrated seamlessly into the existing sensor network, providing additional control input channels that complement automated occupancy-based lighting.

Future integration with classroom electrical systems will be feasible through relay switches that can interface with conventional light fixtures, with the capability to control individual rows of lights based on gesture commands or automated zoning logic. The local database implementation using MySQL allows for storage of historical energy consumption data, lighting schedules, gesture control logs, and system events. Data synchronization between microcontroller units and the centralized dashboard occurs through local network Wi-Fi connections.

6.2.2. Market Feasibility

Market feasibility assesses the demand for LumineSense within educational institutions and evaluates whether the product addresses genuine needs in the target market. Educational facilities face increasing pressure to reduce operational costs and meet sustainability targets, creating a strong market demand for energy management solutions. Many institutions currently rely on manual lighting controls that result in significant energy waste when lights remain on in unoccupied classrooms. LumineSense addresses this market gap by proposing an affordable, scalable IoT solution specifically designed for the

operational patterns and budgetary constraints of schools. The addition of gesture control functionality enhances the system's appeal by offering intuitive, contactless interaction that improves hygiene and accessibility while providing instructors with convenient manual override capabilities during teaching sessions.

The growing emphasis on green campus initiatives and environmental certifications further strengthens the market potential for automated lighting systems. Comparable products in the market either focus on residential applications, whereas LumineSense offers an open-platform solution tailored to educational environments with both automated intelligence and flexible manual control options. The system's ability to demonstrate cost savings through energy consumption reports and daily savings summaries provides clear value propositions that align with institutional procurement priorities and sustainability goals.

6.2.3. Operational Feasibility

The practical features of deploying and sustaining LumineSense within the limitations of team capabilities, available resources, and other operational requirements are assessed by the operational feasibility. This covers the technical and market feasibility sections of feasibility studies mentioned above. Utilizing techniques that expedited the application's development and produced a minimal viable product to showcase and illustrate its essential features, the team plans ahead.

The Cost and Benefits detailed in the project provide a thorough evaluation and analysis of the cost and benefit. The distribution of the project among personnel was ensured by evaluating their skill sets and capacities. In order to improve efficiency,

responsibilities including frontend development, backend integration, and documentation were allocated effectively by balancing out individual abilities in different domains. During the documentation and development stages, processes were streamlined and delays were reduced by using a curated plan to track progress and task modifications.

6.3. Time Management

This section is a crucial part of the project development of LumineSense, ensuring that all features are completed within the allotted completion time. The team prioritized key features and maintained consistent progress throughout the project cycle by breaking the development process up into organized parts. To achieve the project's timeline, time management also helps with task distribution, team milestone tracking, and frequent consultation.

The stages of the development process are used to organize time management. Planning and documentation, core feature implementation, testing, and deployments are the main stages into which these procedures are often divided. For the team members, these stages are significant turning points that guarantee the team won't fall behind on the project schedule.

By implementing a structured time management plan with clear phases and milestones, LumineSense remains a viable product, ensuring a timely delivery while providing the applications core functionalities. This organized approach helps in distributing the teams resources and enables the team to meet the deadline without compromising the functionality of the application. The specific tasks and scheduling activities of the project are described on Appendix B.

6.4. Communication, Coordination, and Team Composition

Successful project management is based on efficient collaboration, communication, and responsibility distribution. These components guarantee that the group stays focused on goals, efficiently handles work, and adjusts to unforeseen difficulties. The team maintained productivity and was able to withstand any issues in the current and future development cycles of the application by ensuring clear communication, clearly defined roles and duties, and strategies.

6.4.1. Communication

To make sure that everyone in the team is aware of the project, task, and priorities, effective communication is crucial. As stated in the sections above, the team made use of communication tools. In addition to managing big files and formal settings in Microsoft Teams, using a straightforward medium like Facebook Messenger and Discord guaranteed that members took part in meetings, check-ups, and documentation cycles. In order to ensure that members are aware of deadlines and tasks assigned to them, the team also attended frequent consultations and in-person meetings for a more thorough work assignment and meetings.

6.4.2. Coordination

Team collaboration is supported by well-defined role assignments, cooperative planning, and ongoing progress monitoring. The team was able to work together on a single project documentation by organizing files, tabs, and outstanding tasks using shared project documents like Google Docs. Each has a thorough explanation, due dates, feedback, and task assignment, all of which serve to distribute the responsibilities among the members.

In addition, coordinated meetings and feedback are important during the project cycles, that ensured the assignments are clearly defined amongst the members, and avoiding miscommunications and disagreements during the timeline and task assignment.

6.4.3. Team Composition

Each of the team's members has a specific assignment position and set of duties. Throughout the development of LumineSense, it was crucial to clearly define team roles and duties in order to ensure accountability, productivity, and a balanced workflow. To facilitate effective work distribution and improved task management, each member was given a distinct assignment based on their knowledge and proficiency with the project's requirements. These positions are intended to handle important project components like development, testing, deployment, and planning. Overall, systematic discussions and the value communication encapsulates the team for the project-based maintenance and actuality wherein in the long term, the project will be finished on time and is well-maintained with regards to benefits in further development.

6.4.3.1.Writer

The responsibility of the writer is essential for creating clear and concise documentation. This role allows developers to program and implement functionalities and designers to blueprint the User Interface (UI) and User Experience (UX). Their responsibility allows other roles to fulfill their task by providing brief and detailed descriptions of functionalities and user guides, allowing for a faster development of the application.

6.4.3.2.Developer

The developer is responsible for the technical backbone of LumineSense, responsible for frontend and backend functionalities. Its responsibility includes creating a structured overview of how different interfaces interact with one another, integrating APIs, and developer tools to be used. In addition, managing the overall logic and database for the application, ensuring the delivery of the applications core features and optimized codebase for rapid prototyping, testing, and deployment.

6.4.3.3.Designer

The designer focuses on making the application's user interface and experience pleasing and user-friendly. This includes producing the wireframes and interactive components needed for the projects, such as posters, icons, and graphical pictures. The application's usability, attractiveness, and accessibility are also guaranteed. They must also continuously work with the developer to make sure that designs are implemented consistently across interfaces and are technically feasible. The success of the application depends on cooperation between roles. While the developer turned the designs into useful functionality, the writer and designer collaborated to incorporate content into a visual layout. The team made sure LumineSense provides a comprehensive, user-focused experience that satisfies both technical and aesthetic requirements by encouraging teamwork.

This chapter's strategic project management structure shows how meticulous planning, precise role description, and methodical collaboration support the successful development of LumineSense. The project builds a strong foundation for delivering a useful and significant energy management solution by carefully examining user classes, evaluating technical and market viability, putting organized time management techniques into place, and encouraging productive team communication.

The comprehensive project management approach ensures that LumineSense achieves its technological goals and meets the practical requirements of educational establishments looking for economical and environmentally friendly lighting solutions. The team has positioned LumineSense to move from concept to implementation while retaining flexibility to adjust to obstacles and take stakeholder feedback into account throughout the development lifecycle through the methodical use of these project management concepts. The operational roadmap that directs LumineSense toward effective deployment and long-term viability in the educational technology landscape is provided by the techniques and tactics covered in this chapter.

Bachelor of Science in
INFORMATION
TECHNOLOGY

GLOSSARY

A. APPLICATION PROGRAMMING INTERFACE (API)

A set of tools that allows different programs or systems to communicate with each other by sending and receiving data, making it possible for the hardware, software components, and web-based dashboard of the system to work together properly.

B. ARDUINO

An open-source electronics platform used for the project. It is designed to be an accessible tool for experts and hobbyists by creating projects using software and hardware programming.

C. CLASSROOM

The physical learning space where LumineSense system is installed where it is designed to monitor occupancy and manage lighting automation. The ideal classroom environment the project focused on is sourced by a national DepEd Order No. 67 stating the ideal dimensions and specifications of a classroom setting, which is specified to be 7 meters in width and 9 meters in length.

D. DAYLIGHT

The general illumination of a given space, whether from natural or artificial sources. In terms of the project, ambient light is used as a measurement of how much the IoT bulb should be dimmed in order to save energy.

E. ENVIRONMENT

It is where the sensors gather their information from. The accuracy of the sensors also relies independently on the information that can be gathered from the environment.

F. IN-CLASS HOURS

Hours wherein a classroom is occupied for a class discussion, whether it be lecture or laboratory hours.

G. INFRARED LIGHT

Infrared light is utilized in LumineSense primarily through Passive Infrared (PIR) sensors to detect motion by sensing changes in infrared radiation emitted from heat-emitting objects such as the human body.

H. INPUT/OUTPUT (I/O)

Process of sending data into a system through devices and receiving results through output devices, allowing the system to interact with users and other devices.

I. INTERNET OF THINGS (IoT)

A network of interconnected devices that communicate, share data, and work together over the internet to perform tasks automatically.

J. LUMINESENSE

The title of the project. 'Lumine,' meaning to give light, and 'sense,' a stimulus. A direct reference to the word luminescence, or an emission of light.

K. MICROCONTROLLER UNIT (MCU)

A microcontroller is a small computer on a single chip that contains a central processing unit (CPU), memory, and input/output (I/O) peripherals. It is designed to perform a specific task within an embedded system, such as controlling a home appliance, a car engine, or an IoT device.

L. PASSIVE INFRARED (PIR) SENSOR

An electronic sensor capable of detecting motion by sensing changes in infrared that is radiated from objects that emit heat. A PIR sensor is used in LumineSense in place of traditional light switches, wherein the light bulb is turned on after detecting body heat and turned off after a set amount of time.

M. SENSORS

A sensor is a device, module, machine, or subsystem that detects events or changes in its environment and sends the information to other electronics, frequently a computer processor.

N. TIMETABLES

A schedule containing in-class hours is typically used in academic settings. A timetable is used in the dashboard to determine when the lighting should be turned on or not.

O. WIRELESS FIDELITY (Wi-Fi)

A wireless networking technology that allows the capability to connect to each other. This technology is used in this project to connect the bulb to the Arduino component where the sensors are residing, allowing lighting automation.

APPENDIX A

Capstone Project Proposal.

Capstone Project Concept Title	Smart Classroom Light Management System		
Concept Author and Collaborator(s)	Ballesteros, Alexandra Shierlmole S.	Entapa, John Gemar E.	Talanquines, Athea Grace O.
	Main Author	Collaborator #1	Collaborator #2

Rationale of the Capstone Project Concept

1. Energy Conservation – Educational institutes waste electricity through lights left on in unoccupied classrooms, which adds to unnecessary energy consumption and increased costs.
2. Environmental Sustainability – Reducing unnecessary energy consumption directly supports environmental conservation efforts by decreasing carbon footprint and promoting sustainable practices within schools.
3. Operational Efficiency – Manual light management can be a hassle to the staff of a large educational institute. An automated system reduces human error and ensures energy management across all classroom spaces.
4. Security Enhancement – The system doubles as a monitoring tool, detecting unauthorized presence during non-operational hours and providing alerts to the security personnel, enhancing overall campus safety.

Main Features and Functionalities

1. Motion Detection and Auto-Control System – PIR sensors detect room occupancy and control lighting with a 10-minute delay before shutdown. This addresses energy waste by eliminating human error and ensuring lights are on if the room is occupied.
2. Facility Management Dashboard – A monitoring system that manages room usage patterns, energy consumption, and system status across all classrooms.
3. Schedule Management System – In the administrative dashboard, authorized personnel can set custom lighting schedules for different rooms based on class timetables.
4. Security System – Night-mode feature that can detect unauthorized presence during designated off-hours and send immediate alerts to the security personnel and admins.

Project Usability / Benefit Justification

This system addresses operational inefficiencies by automating lighting management in a single classroom setting for the meantime. The implementation will provide data for scaling decisions while serving as physical demonstration of IoT applications in educational settings. The security feature enhances campus safety as well. Benefits would include reduced burden on teachers, extended bulb lifespan through optimized usage patterns, and improved environmental sustainability.

Relevance to the Theme

SDG 7 (Affordable and Clean Energy): Promotes energy efficiency and reduces wasteful consumption in educational facilities.

SDG 13 (Climate Action): Directly reduces carbon footprint through decreased energy consumption.

Technology Deployment

Hardware:

- PIR Motion Sensors
- IP Camera (with night vision capability)
- Microcontroller Units (Arduino/Raspberry Pi) – for local processing
- Relay Switches – for light control
- Wireless Communication Modules (Wi-Fi)

Software:

- Web-based dashboard for scheduling, analytics, and system management
- Local Processing Firmware – for sensor integration and automated control
- Local Database – for storing schedules, usage data, and system logs
- Push Notification Services – for security alerts

Anticipated Challenges

1. Testing Environment – Creating a controlled space to simulate classroom conditions.
1. Sensor Calibration – Fine-tuning in a controlled setting rather than live classroom.
2. Hardware Integration – Focus on circuit design and component reliability
3. Software Development – Building and testing the applications and interfaces
4. System Reliability – Ensuring the prototype works consistently during development

References

Hanum, Gilang, et al. 'Integration of Internet of Things (IoT) Technology in the Management of Educational Facilities and Infrastructure'. Al-Fikrah: Jurnal Manajemen Pendidikan, vol. 12, 06 2024, p. 189, https://doi.org/10.31958/jaf.v12i1.13051.

Lopez, Neil Stephen, et al. Energy Audit and Analysis of the Electricity Consumption of an Educational Building in the Philippines for Smart Consumption. 12 2017, pp. 1–4, https://doi.org/10.1109/HNICEM.2017.8269505.

Mojica, Edison. "Evaluation of Factors that Influence Electrical Energy Consumption in Higher Educational Institutions – Preparatory for Energy Management System." E3S Web of Conferences, vol. 433, Oct. 2023, pp. 1–8. https://doi.org/10.1051/e3sconf/202343303004.

Moudgil, Vipul, et al. 'Integration of IoT in Building Energy Infrastructure: A Critical Review on Challenges and Solutions'. Renewable and Sustainable Energy Reviews, vol. 174, no. 113121, Elsevier BV, Mar. 2023, p. 113121, https://doi.org/10.1016/j.rser.2022.113121.

Poyyamozhi, Mukilan, et al. 'IoT—A Promising Solution to Energy Management in Smart Buildings: A Systematic Review, Applications, Barriers, and Future Scope'. Buildings, vol. 14, no. 11, MDPI AG, Oct. 2024, p. 3446, https://doi.org/10.3390/buildings14113446.

APPENDIX B

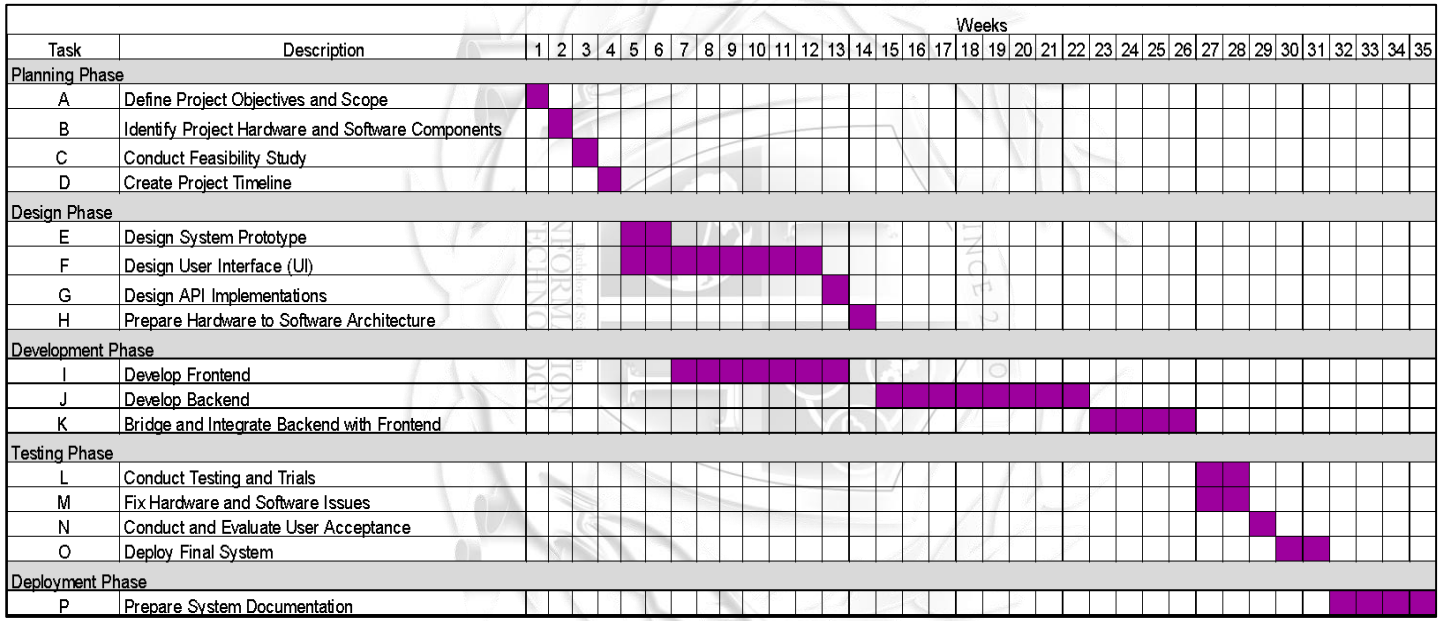
PERT Table.

Task	Description	Predecessor	Sucessor	ET	ES	EF	LS	LF	Slack
Planning Phase									
A	Define Project Objectives and Scope	None	B	1	0	1	0	1	0
B	Identify Project Hardware and Software Compone	A	C	1	1	2	1	2	0
C	Conduct Feasibility Study	B	D	1	2	3	2	3	0
D	Create Project Timeline	C	E,F	1	3	4	3	4	0
Design Phase									
E	Design System Prototype	D	G	2	4	6	10	12	6
F	Design User Interface (UI)	D	G	8	4	12	4	12	0
G	Design API Implementations	E,F	H	1	12	13	12	13	0
H	Prepare Hardware to Software Architecture	G	I,J	1	13	14	13	14	0
Development Phase									
I	Develop Frontend	H	K	7	14	21	15	22	1
J	Develop Backend	H	K	8	14	22	14	22	0
K	Bridge and Integrate Backend with Frontend	I,J	L,M	4	22	26	22	26	0
Testing Phase									
L	Conduct Testing and Trials	K	N	2	26	28	26	28	0
M	Fix Hardware and Software Issues	K	N	2	26	28	26	28	0
N	Conduct and Evaluate User Acceptance	L,M	O	1	28	29	28	29	0
O	Deploy Final System	N	P	2	29	31	29	31	0
Deployment Phase									
P	Prepare System Documentation	O	None	4	31	35	31	35	0

Bachelor of Science in
INFORMATION
TECHNOLOGY

APPENDIX C

Gantt Chart.



APPENDIX D

Minutes of the Capstone Project Proposal Defense.

December 16, 2025 (5:00 PM – 6:30 PM)

College of Information Technology

University of Negros Occidental – Recoletos

Project Proposal Title	LUMINESENSE
Group Members	Alexandra Shierlmole S. Ballesteros John Gemar E. Entapa Athea Grace O. Talanquines
Panel Members	Daniel Von Guillano Joshua Dela Cruz Josiah Frylle Bontilao Elmer T. Haro, Ph.D.
Interpellators	Denro Maban Grace Lynn Javier Francis Louis Villanueva

1. The conference started at the CIT Office Consultation Room with a prayer led by John Gemar E. Entapa.
2. The group members were introduced by Alexandra Shierlmole S. Ballesteros.
3. The oral presentation was delivered by Alexandra Shierlmole S. Ballesteros, covering important proposal details from the essential system features and prototype.
4. After the oral presentation, the following are the suggestions and recommendations made by the interpellators and the panel members:

Chapter or Section	Recommendations or Suggestion	Proponent	Action Taken	Reference or Proof
Title and Front Matters				
Introduction and System Decomposition				
System Features and Functionalities	Power consumption of components especially in Gesture Recognition that contradicts energy conservation.	Denro Maban	A voltage regulator is present to manage power flow over sensors and camera is replaced with host device web camera.	Page 18, Table 2. and Page 20, Table 4.

	Power consumption of IoT components and its feasibility towards the proposed prototype. Concerns of power spiking components contradicting energy conservation.	Joshua Dela Cruz Daniel Von Guillano (follow-up)	Defined power consumption of components and its power sources, which are cross-examined by project expert.	Page 34, 2.3.7.
	Features of user authentication and privilege access. Existence of user access control definitive to various rooms being used by the system and its overall hierarchy.	Daniel Von Guillano Elmer T. Haro, Ph.D. (follow-up)	Added as a software-side feature in the System Decomposition.	Page 41, 3.1. Page 69, 3.2.8.
	Fallback when mistaken or accidental gestures are performed when a lecture is being held. Confirmation gesture to serve as fallback on undesired gesture activation.	Josiah Frylle Bontilao and Joshua Dela Cruz Francis Louis Villanueva (suggestion)	Added a confirmation gesture and dashboard-based popup fallback to clarify gesture performed.	Page 32, 2.3.5.
	Delineating access of gesture controls between users and non-users (e.g.: students)	Grace Lynn Javier	Replacement of IP camera to webcam, which is required by the host device as put into specifications. Confirmation gesture delineates false positives of accidental and undesired gestures.	Page 32, 2.3.5. Page 20, Table 4. Page 32, 2.3.5.
	MediaPipe specifications of predefined gestures in relation to its accuracy in implementation.	Joshua Dela Cruz	Changed predefined gestures to those default “canned gestures” offered by MediaPipe.	Page 32, 2.3.5.

	Dimensions and variability of the classroom the IoT system adheres to.	Elmer T. Haro, Ph.D.	Added as an Implementation Constraint and software-side feature in the System Decomposition.	Page 43, 3.1. and Page 69, 3.2.7.
	Consideration of accuracy thresholds performed by the PIR and web camera.	Elmer T. Haro, Ph.D.	Removed and reiterated into sensor dependencies and mitigations to be performed.	Page 31, 2.3.4.
	Report analytics of energy consumption based on lighting specifications.	Joshua Dela Cruz	Backed with a formula consulted with the project expert implemented as part of web-based monitoring feature	Page 55, 3.2.4. and Page 59, Activity Diagram
	Automation and regulation of system feature to monitor and regulate lighting performance.	Joshua Dela Cruz	Integrated sensors that manage voltage measurements to detect scenarios in lighting issues.	Page 63, 3.2.6 and Page 65, Activity Diagram
System Prototype or Interface	Recommended system layout for deployment in terms of classroom settings. Totality of system prototype be defined and specified aside from the logical connections detailed.	Joshua Dela Cruz Elmer T. Haro, Ph.D.	Added as part of the Design of the system to be implemented. Ideal and variability of classroom settings and physical prototype implemented as part of the Design Constraints.	Page 29 to 30, 2.3.3. Pages 25 to 28, 2.3.1.
Project Management				
Glossary				
Appendices				
References	Citation of the Data Privacy Act in terms of academically sensitive information being present in the classroom.	Daniel Von Guillano	Added into References and further reiterated in System Decomposition.	Page 12, 1.3.4 Page 41, 3.1.

5. After the deliberation of the interpellators and panel members, the proposal is considered accepted with minor revisions in the documentation and in the system prototype.
6. The conference ended at 6:35 PM.

Transcribed by : John Gemar E. Entapa

Noted by : Elmer T. Haro, Ph.D.

SINCE 2010
Bachelor of Science in
INFORMATION
TECHNOLOGY

APPENDIX E

System Logo.

LumineSense, pronounced as loo-me-neh-sens, incorporates two elements that are entirely vital to the study. First, light, symbolized by the left half of the bulb, is a dependent variable that represents the output of the project: the bright success of lessening workload through the means of IoT. Second, smart technological integration, represented by the brain icon on the right half of the bulb, symbolizes the interconnected means of applying embedded systems in everyday objects, even in the light we see for perceiving, observing, and guidance. In addition, two main components surround the bulb: the Arduino MCU and the IP camera, which signifies the automation of lighting behaviors. The cylindrical rectangles symbolize also two features present in LumineSense: the integrated timetable feature (left) and energy analytics saved from automation (right).

APPENDIX F

Team Logo.

The team's name, with BET Emerging Technologies as its full term, symbolizes the researcher's drive towards incorporating modern systems in the common goal for innovation, as expressed in this project which uses embedded systems for quality of use in daily human routines. The logo's dominant color is silver and white, signifying the sophistication and continuing modernity of IT innovations nowadays. Meanwhile, the color black represents the claw marks of a falcon, symbolizing the team's nurtured skills and professionalism they learned from the College. BET also represents the author's initials of their family names, symbolizing continuing camaraderie outside of research projects. With future systems, the team hopes to excel more in ideating more modern technologies to emerge as one that can selflessly support the community around them. In this step, the researchers hope to delineate manual labor towards lighting management.

REFERENCES

{AH2022} Al Farid, F., Hashim, N., Abdullah, J., Bhuiyan, M. R., Shahida Mohd Isa, W. N., Uddin, J., Haque, M. A., & Husen, M. N. (2022). A Structured and Methodological Review on Vision-Based Hand Gesture Recognition System. Journal of imaging, 8(6), 153. https://doi.org/10.3390/jimaging8060153

{AI2023} AutoDesk Instructables. (2023). Design and Implementation of a 3.3V Power Supply Using AMS1117-3.3 Voltage Regulator. AutoDesk Instructables: lorry. https://www.instructables.com/Design-and-Implementation-of-a-33V-Power-Supply-Us/

{AJ2016} AlFaris, F., Juaidi, A., & Manzano-Agugliaro, F. (2016). Improvement of Efficiency Through an Energy Management Program as a Sustainable Practice in Schools. Journal of Cleaner Production, 135, 794–805. https://doi.org/10.1016/j.jclepro.2016.06.172

{AK2020} Andrews, J., Kowsika, M., Vakil, A., & Li, J. (2020). A Motion Induced Passive Infrared (PIR) Sensor for Stationary Human Occupancy Detection. 2020 IEEE/ION Position, Location and Navigation Symposium (PLANS), Portland, Oregon, April 2020, pp. 1295-1304. https://doi.org/10.1109/PLANS46316.2020.9109909

{AR2025} Arduino. (2025). Arduino About. https://www.arduino.cc/en/software/

{BP2021} Brindha, S., Pal, P., Singh, S., Vinod, A., & Mitra, S. K. (2021). Smart Light Management using IoT. I-Manager's Journal on Information Technology, 10(3), 23-32. doi:https://doi.org/10.26634/jit.10.3.16633

{CE2022} Core Electronics. (2022). Core Electronics Logic Level Converter Bidirectional, 4 Channel. GitHub Repository. https://github.com/CoreElectronics/CoreElectronics-Logic-Level-Converter-Bidirectional

{CO2021} Components 101. (2021). 5V Four-Channel Relay Module. https://components101.com/switches/5v-four-channel-relay-module-pinout-features-applications-working-datasheet

{DE2017} Department of Education. (2017). DepEd Order No. 64, s. 2017. Department of Education. https://www.deped.gov.ph/2017/12/15/do-64-s-2017-establishing-the-minimum-performance-standards-and-specifications-for-deped-school-buildings/

{EE2018} Ermac, E. A. (2018). Energy Conservation in Academic Institutions: An Application of Theory of Planned Behavior. Recoletos Multidisciplinary Research Journal, 6(2). https://doi.org/10.32871/rmrj1806.02.02

{EU2021} European Committee of Standardization. (2002). Light and Lighting - Lighting of Work Places - Part 1: Indoor Work Places. Ageta. https://www.ageta.lt/app/webroot/files/uploads/filemanager/File/info/EN_12464-1.pdf

{EF2025} Electronics For You Bureau. (2025). What Is Light Dependent Resistor (LDR)?. ElectronicsForu. https://www.electronicsforu.com/technology-trends/learn-electronics/ldr-light-dependent-resistors-basics

{GA2025} Google AI for Developers. (2025). Gesture Recognition Task Guide. https://ai.google.dev/edge/mediapipe/solutions/vision/gesture_recognizer

{HI2024} Hanum, G. K., Irwin, A., Sari, R. N., Ardi, A., & Lismaryanti, L. (2024). Integration of Internet of Things Technology in the Management of Educational Facilities and Infrastructure. Al-Fikrah: Jurnal Manajemen Pendidikan, 12(1), 189. https://doi.org/10.31958/jaf.v12i1.13051

{JX2025} Javohir, A. & Xojiakbar, M. (2025). Hand Gesture Controlled IoT System. GitHub. https://github.com/notlukas18/hand-gesture-iot

{KK2023} Kishor, K. & Khapre, G. (2023). Energy Management and Optimization in IoT-Based Home Automation Networks. International Research Journal of Modernization in Engineering Technology and Science, 5(6). https://www.irjmets.com/paperdetail.php?paperId=7e9e34c9d2654a95c3568cf5152ac478

{LM2018a} Last Minute Engineers. (2018a). How HC-SR501 PIR Sensor Works & Interface It With Arduino. https://lastminuteengineers.com/pir-sensor-arduino-tutorial/

{LM2018b} Last Minute Engineers. (2018b). Interface DS3231 Precision RTC Module with Arduino. https://lastminuteengineers.com/ds3231-rtc-arduino-tutorial/

{ME2020} Montalbo, F., & Enriquez, E. (2020). An IoT Smart Lighting System for University Classrooms. International Symposium on Educational Technology, 3–7. https://doi.org/10.1109/ISET49818.2020.00011

{MK2016} MaryPraveena, S., Kavitha, A. K., & Kanmani, R. (2016). Energy Conservation in Smart Home using Wireless Sensor Network. International Journal of Trend in Research and Development, 4(1).

https://www.researchgate.net/publication/318029856_Energy_Conservation_in_Smart_Home_Using_Wireless_Sensor_Network

{MR2021} Mischianti, R. (2021, March 6). ESP32 Practical Power Saving: Manage WiFi and CPU – 1. Renzo Mischianti, Electronic 3D Printing and Embedded Programming. https://mischianti.org/esp32-practical-power-saving-manage-wifi-and-cpu-1/

{MS2023} Mohd Alias, M. E., & Salimin, S. (2023). Smart Lighting System for a Classroom using Solar PV and Internet of Things. Evolution in Electrical and Electronic Engineering, 4(1), 128-138. https://penerbit.uthm.edu.my/periodicals/index.php/eeee/article/view/10404

{NA2016} NASDAQ OMX. (2016, January 6). iLuv's Rainbow8™ Smart LED Light Bulb Allows for Affordable, Total Control over Lights from Your Smartphone: Apple HomeKit-certified Smart WiFi Light Bulb Premieres at CES 2016. NASDAQ News Release Distribution Channel. https://www.proquest.com/wire-feeds/iluvs-rainbow8-tm-smart-led-light-bulb-allows/docview/1780816884/se-2?accountid=34542

{NP2012} National Privacy Commission. (2012). Republic Act 10173 – Data Privacy Act of 2012. https://privacy.gov.ph/data-privacy-act/

{PT2025} Pangarkar, T. (2025). Gesture Recognition Statistics. Market.US Scoop. https://scoop.market.us/gesture-recognition-statistics/

{TE2018} The Engineering Projects. (2018) Introduction to Arduino Mega 2560. The Engineering Projects. https://www.theengineeringprojects.com/2018/06/introduction-to-arduino-mega-2560.html

{UN2024} United Nations. (2024). The Sustainable Development Goals Report 2024. New York, USA: UN DESA. https://desapublications.un.org/publications/sustainable-development-goals-report-2024

{VE2025} Verge Sense. (2025). Common Problems with PIR Sensors in Today's Workplace. https://www.vergesense.com/resources/blog/why-pir-sensors-are-no-longer-enough

{VD1985} Vickery, David J. (1985). Norms and Standards of Educational Facilities. United Nations Educational, Scientific, and Cultural Organization Digital Library. https://unesdoc.unesco.org/ark:/48223/pf0000070131#

{VP2025} Vaidyanathan, A., & P., K. (2025). IoT Based Contactless Switch Using Gesture Control. International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI), 1–5. doi:10.1109/ICDSAAI65575.2025.11011594

{VS2023} Vishwakarma, S. (2023). Exploring the Power of Mediapipe and OpenCV: A Hands-on Project for Hand, Pose and Face Detection. Medium. https://medium.com/@sinu2441/exploring-the-power-of-mediapipe-and-opencv-a-hands-on-project-for-hand-pose-and-face-detection-4904ba223641

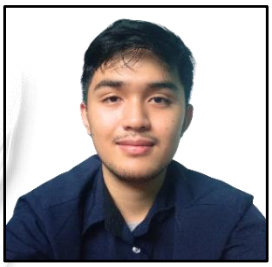
{VS2025} Visual Studio Code. (2025). Visual Studio Code About. https://code.visualstudio.com/docs/editor/whyvscode

{XA2025} XAMPP. (2025). XAMPP About. https://www.apachefriends.org/about.html

SINCE 2010
Bachelor of Science in
INFORMATION
TECHNOLOGY

CURRICULUM VITAE

Name : ALEXANDRA SHIERLMOLE S. BALLESTEROS
ORCID : 0009-0003-4347-1160
Program of Study : Bachelor of Science in Information Technology
Date of Birth : November 8, 2004
Contact Details : 09165093684 / stereoballsgrande@gmail.com
Areas of Interest : Back-End Development, Database Management, Project Management



Name : JOHN GEMAR E. ENTAPA
ORCID : 0009-0009-2208-2063
Program of Study : Bachelor of Science in Information Technology
Date of Birth : August 1, 2004
Contact Details : 09274070202 / johngemare@gmail.com
Areas of Interest : UI/UX Designing, Front-End Development, Project Management

Name : ATHEA GRACE O. TALANQUINES
ORCID : 0009-0007-5670-7645
Program of Study : Bachelor of Science in Information Technology
Date of Birth : February 9, 2005
Contact Details : 09993953867 / talanquines.atheagrace@gmail.com
Areas of Interest : UI/UX Designing, Front-End Development, Project Management